

Mobility Enhancement Techniques

Course content

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- Review of CMOS scaling. Problems with traditional (Dennard) scaling.
- Basic Quantum mechanics and Silicon band structure
- Mobility enhancement techniques (introduced in high volume manufacturing in 2003)
- High k gate dielectrics and metal gates (introduced in 2007)
- FinFETs (introduced in 2012)
- High mobility channel materials (introduced in 2011)
- Nanosheet transistors (introduced in 2024 Samsung Galaxy Watch7)
- Layout dependent effects (became a serious issue since late 2000s)

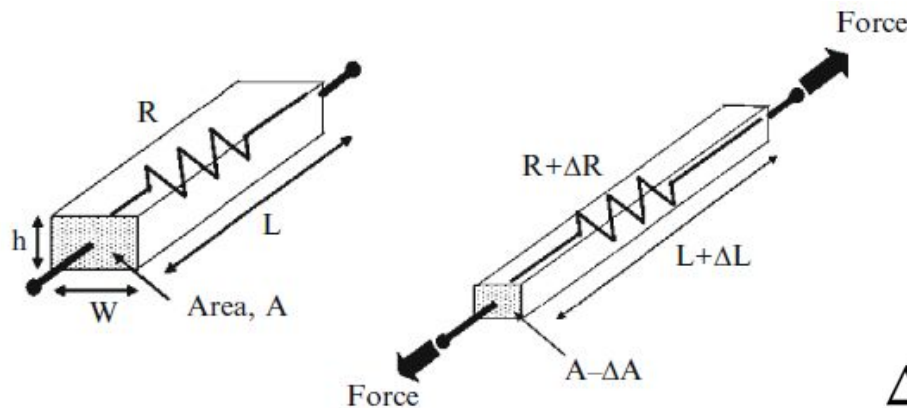
The piezoresistive effect was first reported in 1954 [1] and has been used in making sensors for years. The effect of strain on the mobility of electrons and holes in semiconductors is important in today's sensors and transistors.

1. Charles S. Smith, "Piezoresistance Effect in Germanium and Silicon," *Physical Review*, Vol 94, No.1, April 1, 1954.
2. Y. Kanda, "A graphical representation of the piezoresistance coefficients in silicon," *IEEE Transactions on Electron Devices*, vol. 29, no. 1, pp. 64-70, 1982.

Piezoresistivity

Piezoresistivity is an effect of stress/strain-induced resistivity change of a material.

Piezoresistivity strain sensors are designed to transduce/detect varying strains and produce a proportional change in resistance.



$$\frac{\Delta R}{R} = \frac{\Delta L}{L} - \frac{\Delta W}{W} - \frac{\Delta H}{H} + \frac{\Delta \rho}{\rho}$$

$$\frac{\Delta R}{R} = (1 + 2\nu)\varepsilon_{\text{axial}} + \frac{\Delta \rho}{\rho}$$

$$\frac{\Delta \rho_{ij}}{\rho} = \sum_{k,j} \pi_{ijkl} \tau_{kl} \quad \frac{\Delta \rho_i}{\rho} = \sum_{k=1}^6 \pi_{ik} \tau_k$$

$$\pi_{ik} = \begin{pmatrix} \pi_{11} & \pi_{12} & \pi_{12} & 0 & 0 & 0 \\ \pi_{12} & \pi_{11} & \pi_{12} & 0 & 0 & 0 \\ \pi_{12} & \pi_{12} & \pi_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & \pi_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & \pi_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & \pi_{44} \end{pmatrix}$$

Piezoresistance tensor (rank 4) in matrix form (only 3 indpt entries for Si)

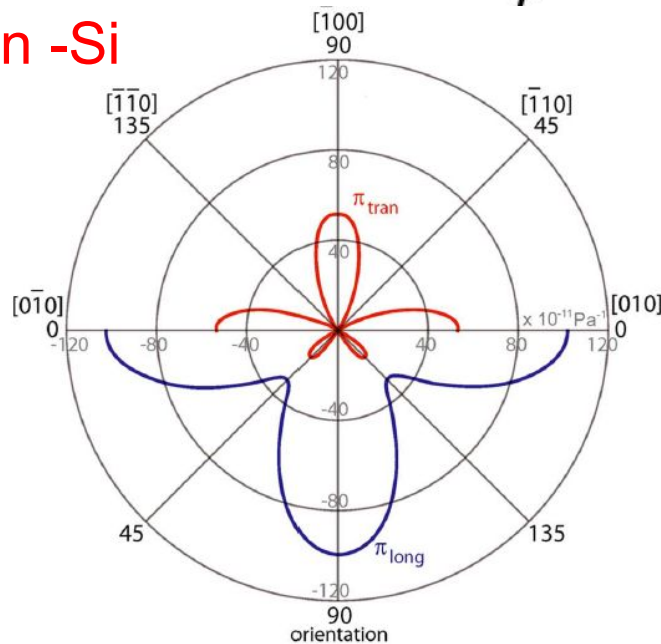
Piezoresistivity

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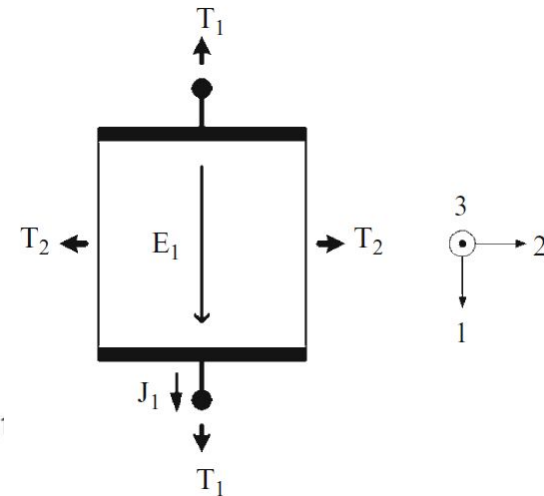
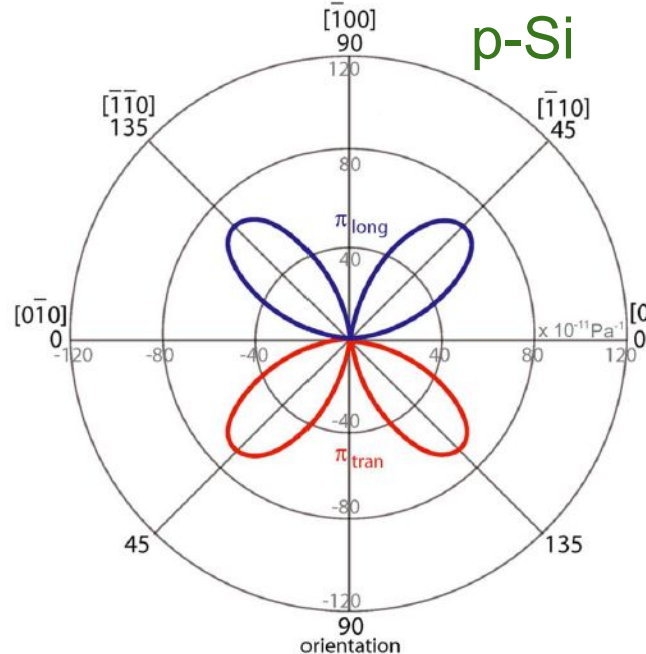
Consider the special case where the stress T_1 is longitudinal with respect to the current density, J_1 , π_{11} may be considered as the longitudinal coefficient, π_L . The stress T_2 is transverse with respect to the current density, J_1 , so π_{12} may be considered as the transverse coefficient, π_T .

$$\frac{\Delta\rho_1}{\rho} = \pi_L T_1 + \pi_T T_2$$

n-Si



p-Si



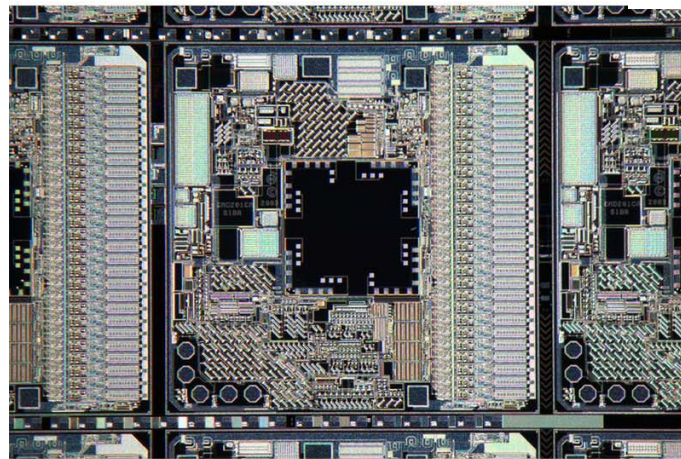
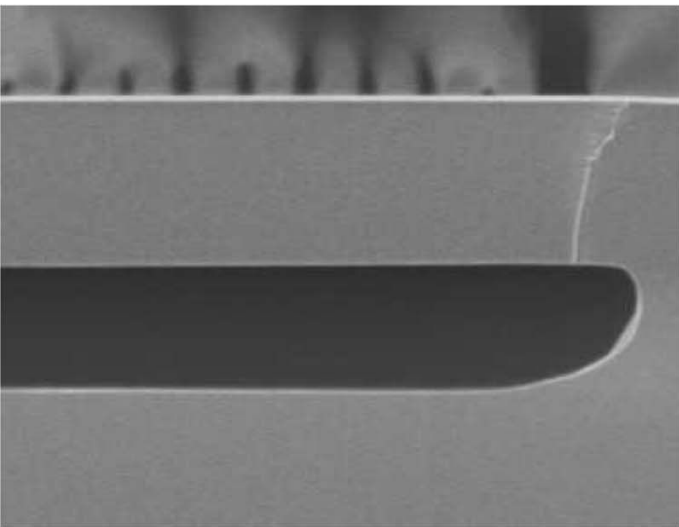
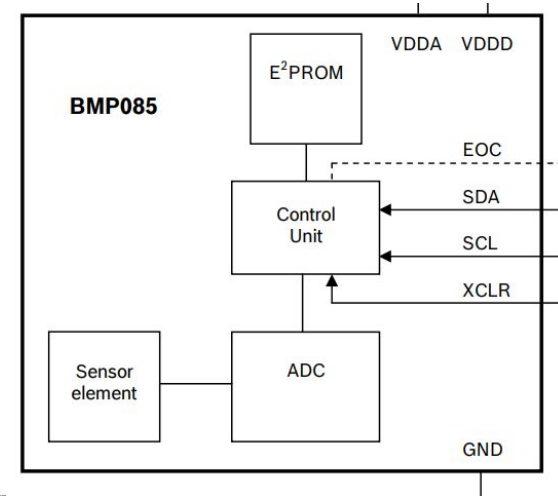
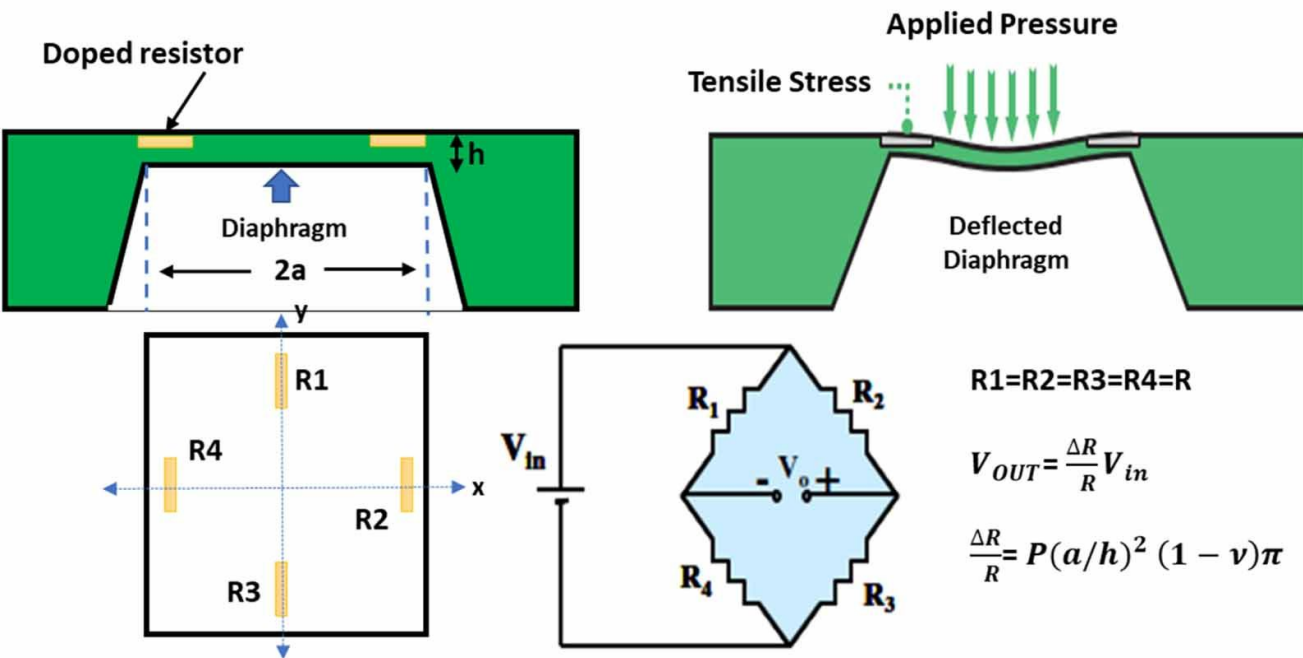
How much stress is needed to reduce the resistance ($\Delta\rho/\rho$) in $\langle 110 \rangle$ direction by 30% for n & p type?

π_L and π_T values (in units of 10^{-11} Pa^{-1}) for (001) Si. Upper half means +ve and lower half means -ve values. Tensile stress - positive and Compressive stress - negative.

[K. E. Petersen, "Silicon as a mechanical material," in *Proceedings of the IEEE*, May 1982](#)

[A. A. Barlian et al, "Review: Semiconductor Piezoresistance for Microsystems," in *Proceedings of the IEEE*, March 2009](#)

Piezoresistive pressure sensor



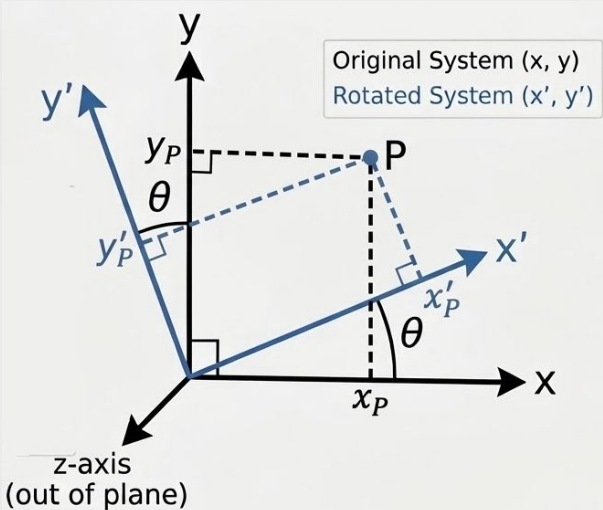
Bosch BMP085
5 mm x 5 mm x 1.2 mm

[K. E. Petersen, "Silicon as a mechanical material," in *Proceedings of the IEEE*, May 1982](#)

[A. A. Barlian et al, "Review: Semiconductor Piezoresistance for Microsystems," in *Proceedings of the IEEE*, March 2009](#)

COORDINATE ROTATION: SINGLE VS. PRODUCT TRANSFORMATION

RANK-1 VECTOR TRANSFORMATION



Coordinates of point P: (x, y) in original axes, (x', y') in rotated axes.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = R(\theta) \begin{bmatrix} x \\ y \end{bmatrix} \quad \begin{bmatrix} x \\ y \end{bmatrix}^T = R(-\theta) \begin{bmatrix} x' \\ y' \end{bmatrix}^T$$

where $R(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ and $R(-\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

$$\begin{aligned} x &= x' \cos\theta - y' \sin\theta \\ y &= x' \sin\theta + y' \cos\theta \end{aligned}$$

STEP-BY-STEP PROOF: PRODUCTS ARE RANK-2 TENSORS

STEP 1: Define Coordinate Product Tensor

$$T = \begin{bmatrix} x^2 & xy \\ xy & y^2 \end{bmatrix}$$

Definition for symmetric matrix

STEP 2: State Tensor similarity transformation

$$T' = R(\theta) T R^T(\theta)$$

T' in primed variables. The components of T' are x'^2, x'y', and y'^2.

STEP 3: Perform Matrix Multiplication

$$T' = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x^2 & xy \\ xy & y^2 \end{bmatrix} \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$RT = \begin{bmatrix} x^2 \cos\theta + xy \sin\theta & xy \cos\theta + y^2 \sin\theta \\ -x^2 \sin\theta + xy \cos\theta & -xy \sin\theta + y^2 \cos\theta \end{bmatrix}$$

$$(RT)R^T = \begin{bmatrix} x^2 \cos^2\theta + y^2 \sin^2\theta + 2xy \sin\theta \cos\theta & -(x^2 - y^2) \sin\theta \cos\theta + xy(\cos^2\theta - \sin^2\theta) \\ -(x^2 - y^2) \sin\theta \cos\theta + xy(\cos^2\theta - \sin^2\theta) & x^2 \sin^2\theta + y^2 \cos^2\theta - 2xy \sin\theta \cos\theta \end{bmatrix}$$

STEP 4: Extract resulting T' Matrix

$$T' = \begin{bmatrix} x^2 \cos^2\theta + y^2 \sin^2\theta + xy \sin(2\theta) & xy \cos(2\theta) - \frac{1}{2}(x^2 - y^2) \sin(2\theta) \\ xy \cos(2\theta) - \frac{1}{2}(x^2 - y^2) \sin(2\theta) & x^2 \sin^2\theta + y^2 \cos^2\theta - xy \sin(2\theta) \end{bmatrix}$$

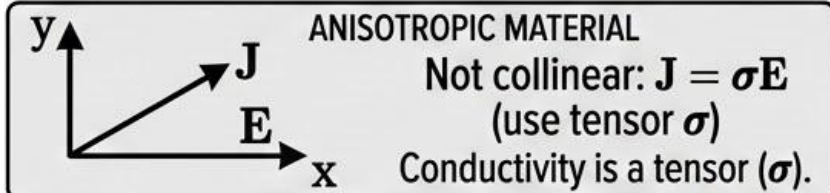
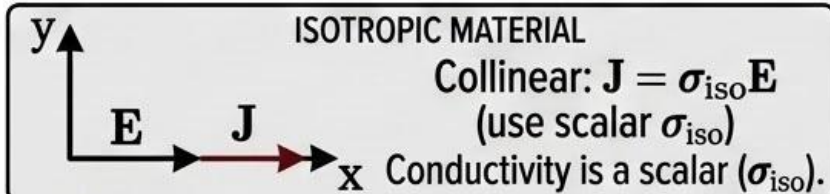
STEP 5: Term mixing and 2θ frequency transformation ofirm Rank-2 nature.

Feature	Rank-1 Vector (x', y')	Product of Two R(θ) • R ^T (Rank-2)
Row 1: Matrix Representation	One R(θ)	Product of Two R(θ) • R ^T
Row 2: Frequency	θ	2θ

PROVING CONDUCTIVITY (σ) IS A RANK-2 TENSOR FROM OHM'S LAW ($\mathbf{J} = \sigma \mathbf{E}$)

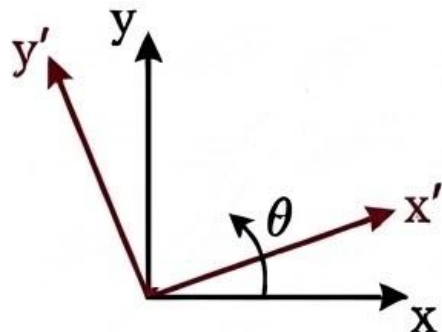
1. SYSTEM DEFINITIONS

1A. Ohm's Law: Isotropic vs. Anisotropic Material



Where $\mathbf{J}$ and $\mathbf{E}$ are Rank-1 Tensors (Vectors)

1B. Coordinate Rotation Transformation ($\mathbf{R}$)



$$\mathbf{J}' = \mathbf{R} \mathbf{J}$$
$$\mathbf{E}' = \mathbf{R} \mathbf{E}$$

$$\mathbf{J} = \mathbf{R}^T \mathbf{J}'$$
$$\mathbf{E} = \mathbf{R}^T \mathbf{E}'$$

2. DERIVATION STEPS

3. STEP 2.1: Substitute transformations

Substitute $\mathbf{J}$ and $\mathbf{E}$ from (1B) into Ohm's Law (1A)

$$\mathbf{R}^T \mathbf{J}' = \sigma (\mathbf{R}^T \mathbf{E}')$$

4. STEP 2.2: Isolate transformed current density ($\mathbf{J}$)

Multiply left side by rotation matrix $\mathbf{R}$

$$\mathbf{R}(\mathbf{R}^T \mathbf{J}') = \mathbf{R} \sigma (\mathbf{R}^T \mathbf{E}')$$

$$\mathbf{J}' = (\mathbf{R} \sigma \mathbf{R}^T) \mathbf{E}'$$

(Since $\mathbf{R} \mathbf{R}^T = \mathbf{I}$ Identity matrix)

3. COMPARISON & CONCLUSION

Principle of Covariance:

Ohm's Law must keep form in new frame:

$$\mathbf{J}' = \sigma' \mathbf{E}' \longrightarrow \sigma' = \mathbf{R} \sigma \mathbf{R}^T$$

A mathematical entity that transforms as

$\mathbf{T}' = \mathbf{R} \mathbf{T} \mathbf{R}^T$ is a Rank-2 Tensor.

Conductivity (σ) follows this rule and is thus proven.

This similarity transformation ensures consistency of the physical law across all frames.

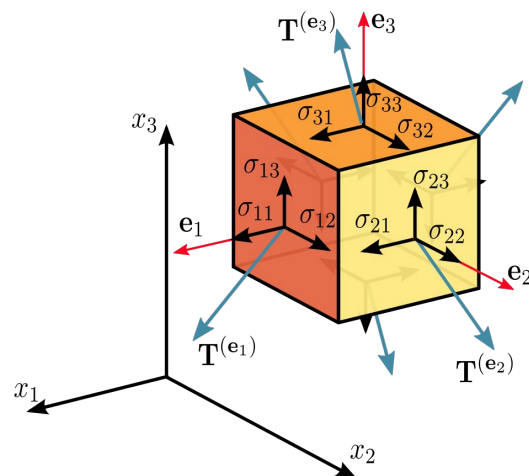
Tensor vs Matrix

- Anisotropy- physical property of a crystal depends on the direction in which it is measured
- All crystal are anisotropic for some of their properties
- Tensors of different ranks are used to represent such properties
- A tensor of rank r in 3-D space has 3^r components. Each component has r suffixes
- The components of a rank 1 tensor (vector) change under a rotation of axes as, $X'_i = R_{ij}X_j$, where R is the 3×3 matrix of direction cosines

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$
- The transformation law for a tensor of rank 2 is $T'_{ij} = R_{ik}R_{jl}T_{kl}$ or in matrix notation $T' = RTR^T$
- A second rank Tensor component T_{ij} transforms like the product $x_i x_j$
- A r rank Tensor component transforms like the product $x_1 x_2 \dots x_r$
- Tensor may be represented by a matrix, but not all matrices are tensors. Matrix is mathematical idea. Any set of numbers can be arranged in a rectangular array. A tensor is a physical concept
- **Tensor in ML/DS/AI is another name for a multidimensional array.**

Stress

- Consider a unit cube within the body. The force transmitted across each face are resolved into three components
- σ_{12} is the stress on the face normal to x_1 direction in the $+x_2$ direction by the material outside the cube upon the material inside it.
- If the stress is homogeneous, the forces exerted on the cube on the opposite faces must be equal and opposite
- $\sigma_{11}, \sigma_{22}, \sigma_{33}$ are normal and $\sigma_{12}, \sigma_{21}, \sigma_{31}, \sigma_{13}, \sigma_{23}, \sigma_{32}$ are shear components
- For pressure, the force acting on the plane is always normal to the plane, whereas with stress the force can be inclined to the plane normal and even in the plane.



Tensile stress - positive

Compressive stress - negative

For static equilibrium with
no torque, $\sigma_{ij} = \sigma_{ji}$

$$\begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix}$$

Stress tensor (rank 2) can be
written as

Types of Stress

Uniaxial

$$\begin{bmatrix} \sigma & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Biaxial

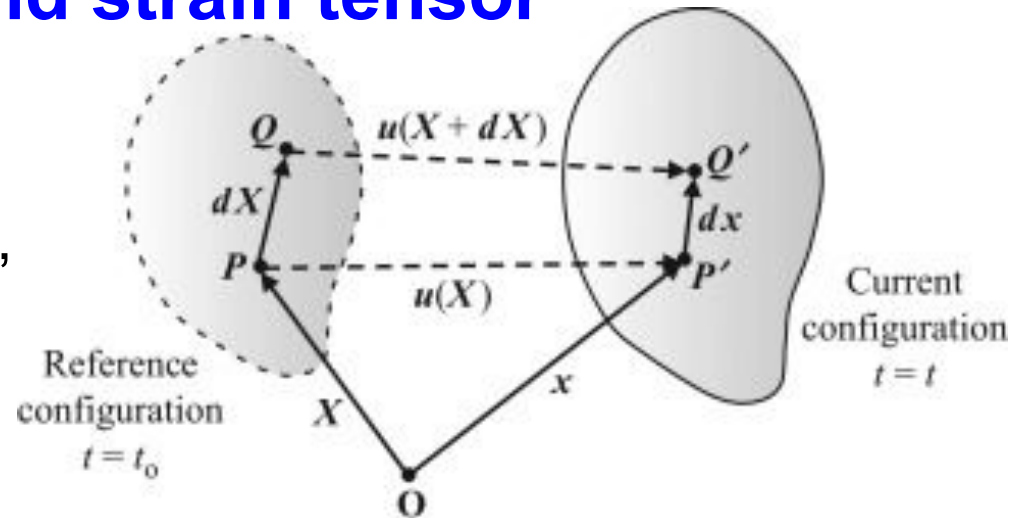
$$\begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hydrostatic pressure

$$\begin{bmatrix} -p & 0 & 0 \\ 0 & -p & 0 \\ 0 & 0 & -p \end{bmatrix}$$

Displacement gradient and strain tensor

Suppose X and $X+dX$ become x and $x+dx$ in the deformed state. We may express $x(X)$ as $X+u(X)$, where $u(X)$ is the displacement undergone by a point at X in the undeformed state



The displacement gradient tensor that describes the local variation of the displacement field within a material can be written as sum of symmetric and anti-symmetric part

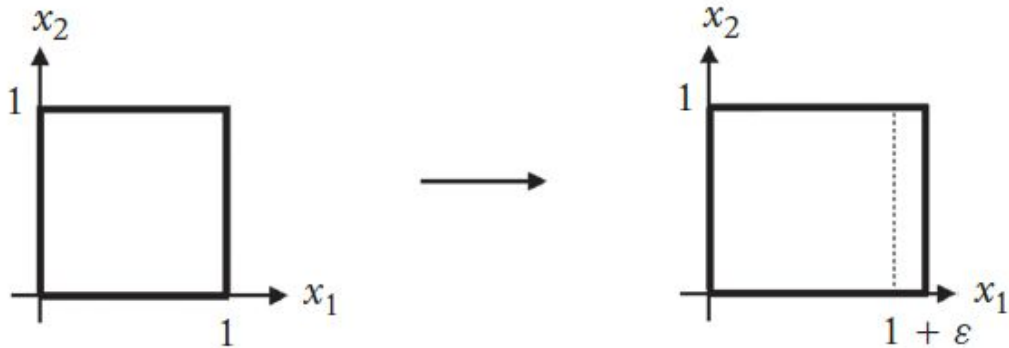
$$\begin{bmatrix} \frac{\partial u_x}{\partial x} & \frac{\partial u_x}{\partial y} & \frac{\partial u_x}{\partial z} \\ \frac{\partial u_y}{\partial x} & \frac{\partial u_y}{\partial y} & \frac{\partial u_y}{\partial z} \\ \frac{\partial u_z}{\partial x} & \frac{\partial u_z}{\partial y} & \frac{\partial u_z}{\partial z} \end{bmatrix} = \varepsilon_{kj} + \omega_{kj}$$

$$\varepsilon_{kj} = \frac{1}{2} \left(\frac{\partial u_k}{\partial X_j} + \frac{\partial u_j}{\partial X_k} \right) \quad \omega_{kj} = \frac{1}{2} \left(\frac{\partial u_k}{\partial X_j} - \frac{\partial u_j}{\partial X_k} \right)$$

Strain tensor

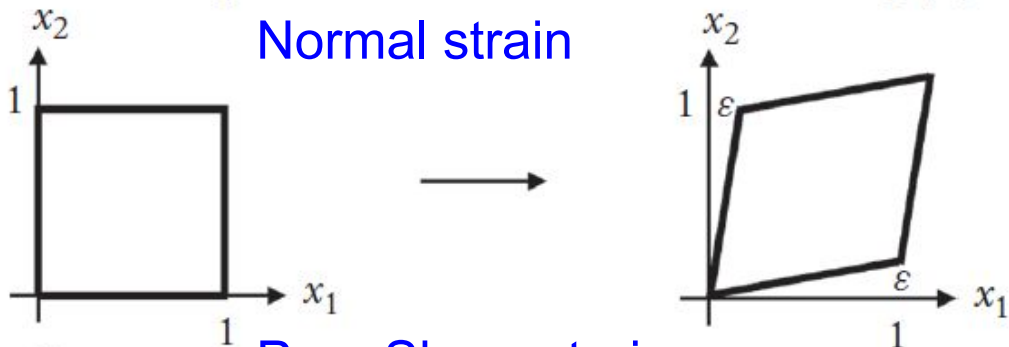
Rotation tensor

Types of Strain



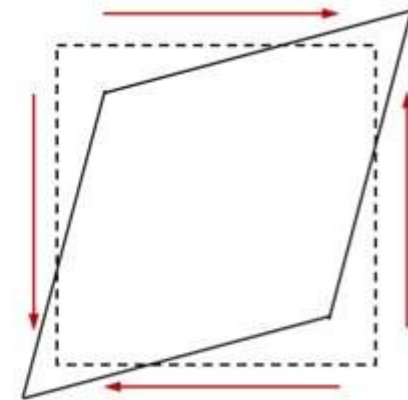
Normal strain

$$\mathbf{e} = \begin{pmatrix} \varepsilon & 0 \\ 0 & 0 \end{pmatrix}$$

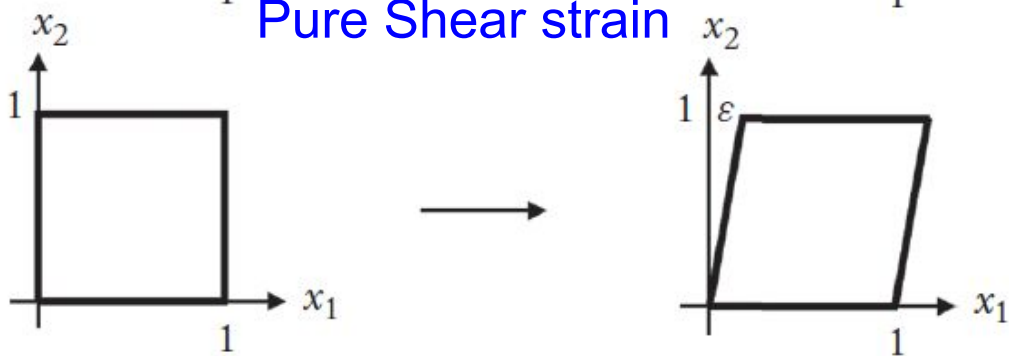


Pure Shear strain

$$\mathbf{e} = \begin{pmatrix} 0 & \varepsilon \\ \varepsilon & 0 \end{pmatrix}$$



Pure Shear



Simple shear

$$\begin{pmatrix} 0 & \varepsilon/2 \\ \varepsilon/2 & 0 \end{pmatrix} + \begin{pmatrix} 0 & \varepsilon/2 \\ -\varepsilon/2 & 0 \end{pmatrix}$$

pure shear strain of $\varepsilon/2$ and a rotation by $\theta = \varepsilon/2$

Does Stress cause Strain or vice versa?

Stress does not cause strain (nor does strain cause stress), but the two are coupled to each other, the presence of either necessarily implies that the other is also present.

Internal deformations for example, can be “caused” by thermal gradients, dislocations, and defects in the crystal lattice or by the presence of dopant atoms that are significantly larger or smaller than the host atoms and thus deform the lattice structure. In such cases, internal forces are established, and it would be proper to refer to these stresses as being the result of the strains.

A further distinction between stress and force comes from the fact that stresses always occur in opposite (but not always equal) pairs. In the static case, the stresses are equal because there is no net motion of any internal volumes. In the dynamic case (e.g., the propagation of an acoustic wave), the opposite stresses are not generally equal.

Generalized Hooke's law

- In linear elasticity theory (displacement gradient $\ll 1$), the strain induced in a medium is proportional to the applied stress and vice versa. The constants of proportionality can be expressed as a fourth rank tensor

$$\varepsilon_{ij} = S_{ijkl} \sigma_{kl}$$

$$\sigma_{ij} = C_{ijkl} \varepsilon_{kl}$$

S_{ijkl} is the elastic compliance tensor C_{ijkl} is the elastic stiffness tensor

Though it appears at first sight that there are $3^4 = 81$ independent elastic constants, crystal symmetry make it a smaller number. The smallest number of independent elastic constants is just two (elastically isotropic material like rubber). In cubic crystals (Si, Ge) there are just three independent elastic constants and in hexagonal crystals five. The largest number of independent elastic constants in any material is 21. This is the case in a triclinic crystal where there are no rotational symmetries in the point group.

Generalized Hooke's law for Cubic materials

For cubic crystals can be expressed in a convenient matrix form

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \\ \sigma_5 \\ \sigma_6 \end{bmatrix} = \begin{bmatrix} c_{11} & c_{12} & c_{12} & 0 & 0 & 0 \\ c_{12} & c_{11} & c_{12} & 0 & 0 & 0 \\ c_{12} & c_{12} & c_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & c_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & c_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & c_{44} \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \\ \varepsilon_6 \end{bmatrix}$$

$$S_{11} = \frac{C_{11} + C_{12}}{(C_{11} - C_{12})(C_{11} + 2C_{12})}$$

$$S_{12} = \frac{-C_{12}}{(C_{11} - C_{12})(C_{11} + 2C_{12})}$$

$$S_{44} = \frac{1}{C_{44}}$$

SIMPLIFIED SUBSCRIPTS FOR CUBIC SYMMETRY

Face/Direction	Condensed Numeric Equivalent
xx or 11	1
yy or 22	2
zz or 33	3
yz or 23	4
zx or 31	5
xy or 12	6

$$\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \\ \varepsilon_6 \end{bmatrix} = \begin{bmatrix} s_{11} & s_{12} & s_{12} & 0 & 0 & 0 \\ s_{12} & s_{11} & s_{12} & 0 & 0 & 0 \\ s_{12} & s_{12} & s_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & s_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & s_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & s_{44} \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \\ \sigma_5 \\ \sigma_6 \end{bmatrix}$$

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \varepsilon_{yz} \\ \varepsilon_{zx} \\ \varepsilon_{xy} \end{bmatrix} = \begin{bmatrix} \frac{1}{E_x} & -\frac{\nu_{yx}}{E_y} & -\frac{\nu_{zx}}{E_z} & 0 & 0 & 0 \\ -\frac{\nu_{xy}}{E_x} & \frac{1}{E_y} & -\frac{\nu_{zy}}{E_z} & 0 & 0 & 0 \\ -\frac{\nu_{xz}}{E_x} & -\frac{\nu_{yz}}{E_y} & \frac{1}{E_z} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{yz}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{zx}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{xy}} \end{bmatrix} \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \sigma_{yz} \\ \sigma_{zx} \\ \sigma_{xy} \end{bmatrix}$$

S_{ij} coefficients are converted to Young's modulus (E), Poisson's ratio (ν), and the shear modulus (G)

Generalized Hooke's law for Silicon

The Young's modulus, Poisson's ratio and Shear modulus of Silicon varies with crystal orientation. The matrix is given for two important directions.

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \\ \sigma_5 \\ \sigma_6 \end{bmatrix} = \begin{bmatrix} 165.7 & 63.9 & 63.9 & 0 & 0 & 0 \\ 63.9 & 165.7 & 63.9 & 0 & 0 & 0 \\ 63.9 & 63.9 & 165.7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 79.6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 79.6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 79.6 \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \\ \varepsilon_6 \end{bmatrix} \text{ GPa}$$

Cubic Silicon ([100], [010], [001]). (x,y,z directions)

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \sigma_4 \\ \sigma_5 \\ \sigma_6 \end{bmatrix} = \begin{bmatrix} 194.5 & 35.7 & 64.1 & 0 & 0 & 0 \\ 35.7 & 194.5 & 64.1 & 0 & 0 & 0 \\ 64.1 & 64.1 & 165.7 & 0 & 0 & 0 \\ 0 & 0 & 0 & 79.6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 79.6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 50.9 \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \varepsilon_4 \\ \varepsilon_5 \\ \varepsilon_6 \end{bmatrix} \text{ GPa}$$

Standard (001) Silicon Wafer ([110], [$\bar{1}10$], [001]). (x,y,z directions)

Young's modulus (E), Poisson's ratio (ν), and the shear modulus (G) in the axes of interest (x, y, z);

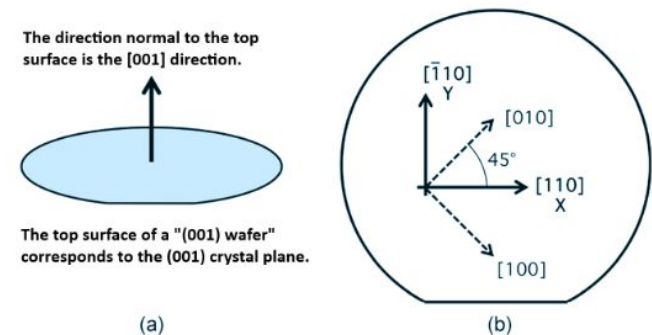


Fig. 2: Crystal orientation of a standard (001) silicon wafer. (a) The wafer surface is aligned with the (001) plane. (b) The in-plane X- and Y-axes are oriented along the [110] and [$\bar{1}10$] directions, respectively.

Isotropic materials (eg: Tungsten)

$$\varepsilon_1 = \frac{\sigma_1}{Y} - \frac{\nu\sigma_2}{Y} - \frac{\nu\sigma_3}{Y}$$

$$\varepsilon_4 = \frac{2(1+\nu)}{Y}\sigma_4$$

$$\varepsilon_2 = \frac{\sigma_2}{Y} - \frac{\nu\sigma_1}{Y} - \frac{\nu\sigma_3}{Y}$$

$$\varepsilon_5 = \frac{2(1+\nu)}{Y}\sigma_5$$

$$\varepsilon_3 = \frac{\sigma_3}{Y} - \frac{\nu\sigma_1}{Y} - \frac{\nu\sigma_2}{Y}$$

$$\varepsilon_6 = \frac{2(1+\nu)}{Y}\sigma_6$$

Consider a tensile test where a sample is loaded in tension along the x_3 axis and no constraints or loads are applied along x_1 and x_2 ($\sigma_1=\sigma_2=0$)

$$\nu = -\frac{\varepsilon_1}{\varepsilon_3} = -\frac{\varepsilon_2}{\varepsilon_3} \quad Y = \frac{1}{S_{11}}, \quad \nu = -\frac{S_{12}}{S_{11}}, \quad G = \frac{Y}{2(1+\nu)}$$

For isotropic materials, only two parameters are independent.

Poisson's ratio ($-1 \leq \nu \leq 1/2$) is the ratio of the contraction in the lateral x_1 and x_2 directions to the tensile strain along x_3 .

Try this problem

An accelerometer which works on the principle of piezoresistance is shown below. It consists of a Wheatstone bridge placed near the fixed end of a cantilever made in a Silicon wafer. In the Wheatstone bridge, two are Platinum resistors (P_X, P_Y) and rest are of p-type Silicon (S_X, S_Y). All four are rectangular with finite length, width and height. The value of all four resistances are same when there is no acceleration. A proof mass is connected to the free end of the cantilever. Under some force, the proof mass end deflects and creates a tensile stress (+3.3 GPa) along [110] direction at the fixed end. Find the longitudinal and transverse (in-plane and z direction) stress (τ), strain (ε), relative change in resistance ($\frac{\Delta R}{R}$) due to change in geometry and piezoresistive effect for each resistor. Then find the net V_{out}/V_{in} . Neglect the effect of any thermal expansion due to heating and effect of stress on interconnecting wires.

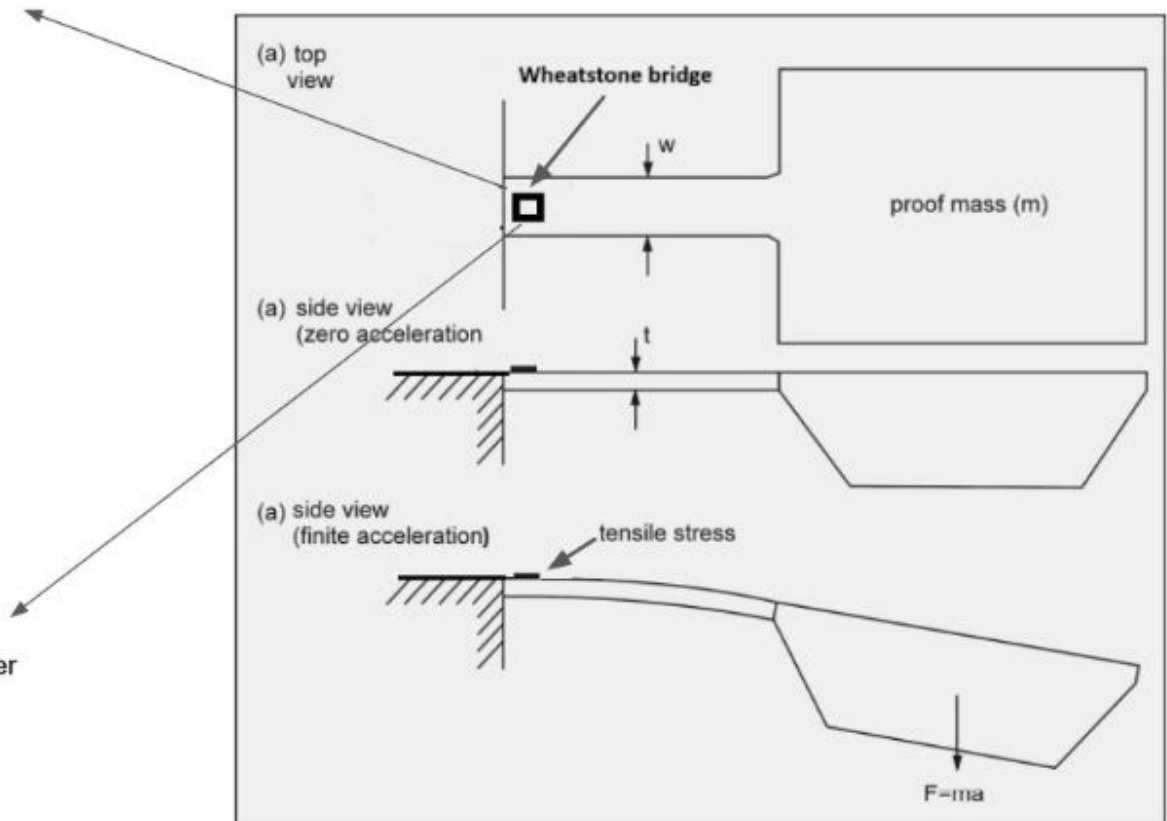
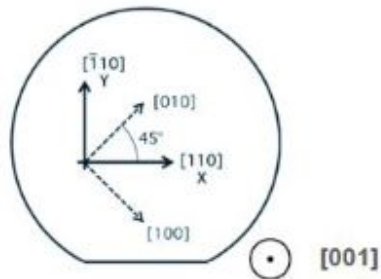
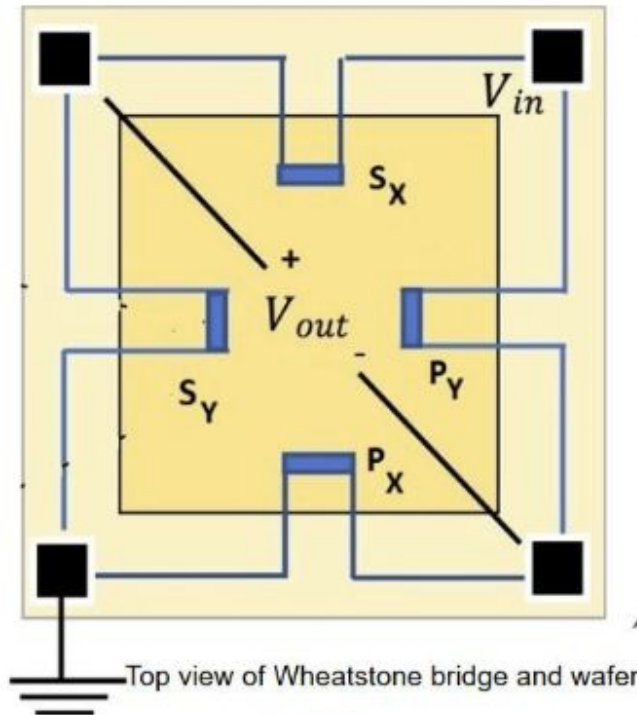
Resistor	$\tau_L(GPa), \varepsilon_L$	$\tau_T(GPa), \varepsilon_T$	$\tau_{zz}(GPa), \varepsilon_{zz}$	$\frac{\Delta R}{R}$ (Geometry)	$\frac{\Delta R}{R}$ (Piezo)	Net $\frac{\Delta R}{R}$
P_X						
P_Y						
S_X						
S_Y						

V_{out}/V_{in} is

The elastic (Young's modulus and Poisson's ratio) and piezoresistance parameters of Si and Pt are

$$Y^{Pt}=165 \text{ GPa}, \quad \nu^{Pt}=0.35, \quad \pi_L^{Pt} = 0, \quad \pi_T^{Pt} = 0$$

$$Y^{Si}=165 \text{ GPa}, \quad \nu^{Si}=0.35, \quad \pi_L^{Si} = 70 \times 10^{-11} \text{ Pa}^{-1}, \quad \pi_T^{Si} = -70 \times 10^{-11} \text{ Pa}^{-1}$$



Strain Effects on the Silicon Band Structure

- The deformation potential theory relates the shifts of the CB energy bands to small deformations of the crystal

$$\Delta E(\mathbf{k}) = \sum_{ij} D_{ij}^{(\mathbf{k})} \varepsilon_{ji}$$

- In case of bands degenerate at a certain symmetry point of the Brillouin zone strain causes not only relative shifts between the bands but also modifies their shapes. In order to describe the band warping, an additional Hamiltonian linear in strain is added to a k.p Hamiltonian of the relaxed crystal.

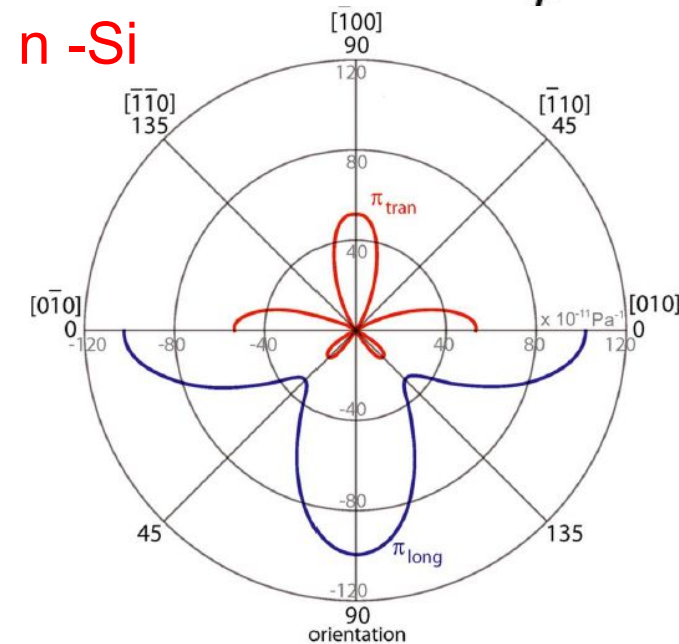
$$\delta H(\hat{\varepsilon})_{mm'} = \sum_{i,j} D_{mm'}^{ji} \varepsilon_{ij},$$

$$H(\hat{\varepsilon}) = H + \delta \mathbf{H}_{strain} + H_{so}$$

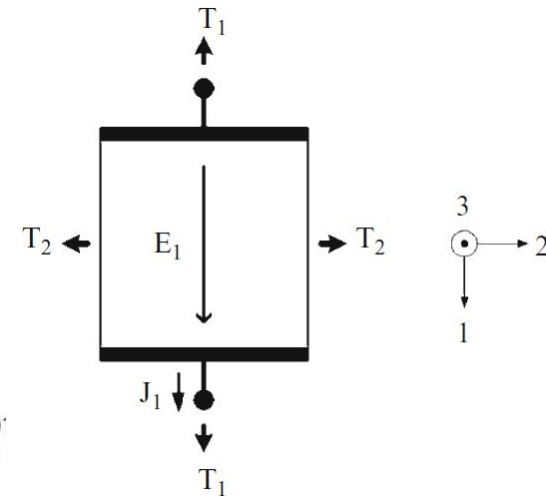
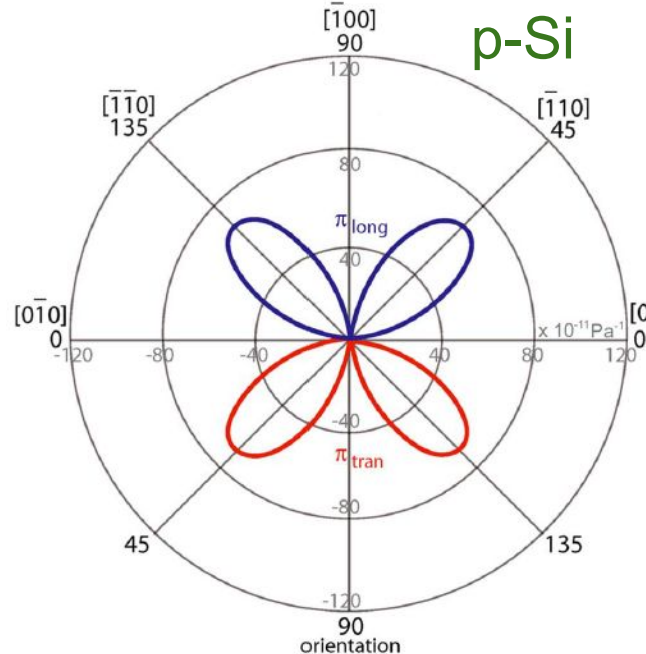
Consider the special case where the stress T_1 is longitudinal with respect to the current density, J_1 , π_{11} may be considered as the longitudinal coefficient, π_L . The stress T_2 is transverse with respect to the current density, J_1 , so π_{12} may be considered as the transverse coefficient, π_T .

$$\frac{\Delta\rho_1}{\rho} = \pi_L T_1 + \pi_T T_2$$

n-Si



p-Si



How much stress is needed to reduce the resistance ($\Delta\rho/\rho$) in $\langle 110 \rangle$ direction by 30% for n & p type?

π_L and π_T values (in units of 10^{-11} Pa^{-1}) for (001) Si. Upper half means +ve and lower half means -ve values. Tensile stress - positive and Compressive stress - negative.

[K. E. Petersen, "Silicon as a mechanical material," in *Proceedings of the IEEE*, May 1982](#)

[A. A. Barlian et al, "Review: Semiconductor Piezoresistance for Microsystems," in *Proceedings of the IEEE*, March 2009](#)

In the $\langle 110 \rangle$ direction

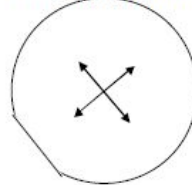
	π_L (E^{-11}/Pa)	π_T (E^{-11}/Pa)
Electrons	-31.6	-17.6
holes	71.8	-66.3

In the $\langle 100 \rangle$ direction

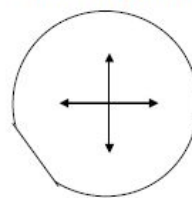
	π_L (E^{-11}/Pa)	π_T (E^{-11}/Pa)
Electrons	-102	53.4
holes	6.6	-1.1

Direction of
Carrier Flow

(100) wafer
 $\langle 110 \rangle$ directions



(100) wafer
 $\langle 100 \rangle$ directions



110 direction on 001 surface

	nMOS Π	pMOS Π
x	3.1%	-7.2%
y	-5.3%	0.1%
z	1.7%	6.6%

Values are mobility change per 100MPa

Signs of Stress

Compressive = (-ve)

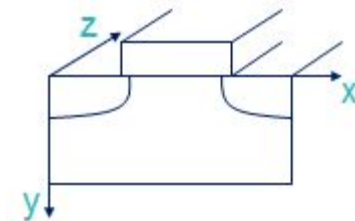
Tensile = (+ve)

■ NFET improvement

- ❑ Tensile X and Z stress
- ❑ Compressive Y stress ❑ most sensitive

■ PFET improvement

- ❑ Compressive X stress ❑ most sensitive
- ❑ Tensile Y and Z stress



Band structure engineering

Mechanical Stress

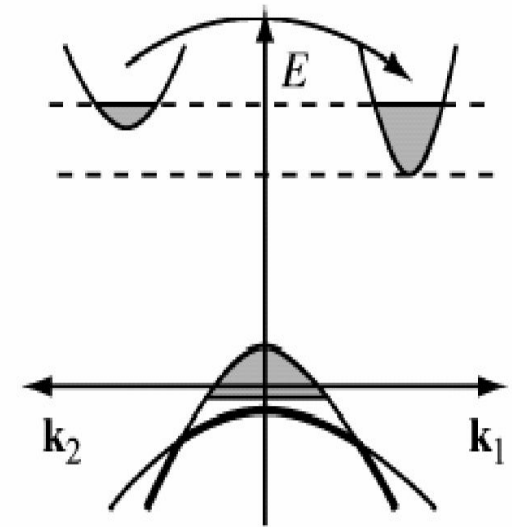
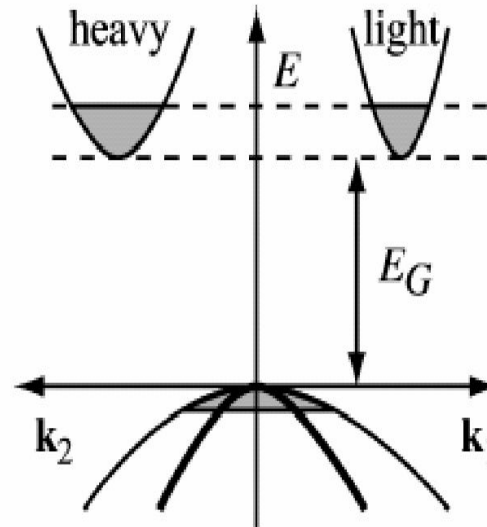
↓
deforms lattice and changes
band structure

↓
Lifts the degeneracy of CB
bottom and VB maximum,
warps/shifts bands

↓
Puts more carriers in a low
effective mass valley/large DOS

↓
less intervalley scattering with
fewer valleys

↓
better mobility



Type of stress

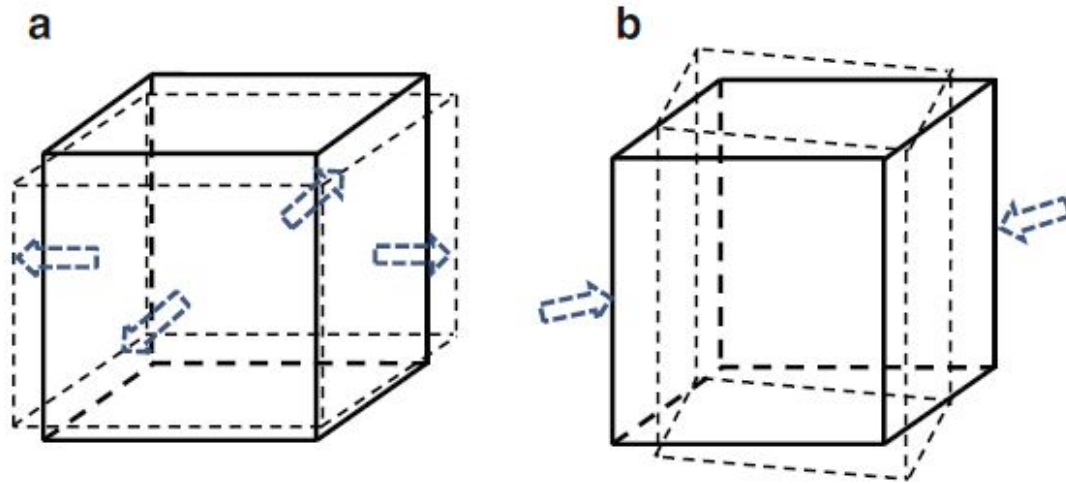
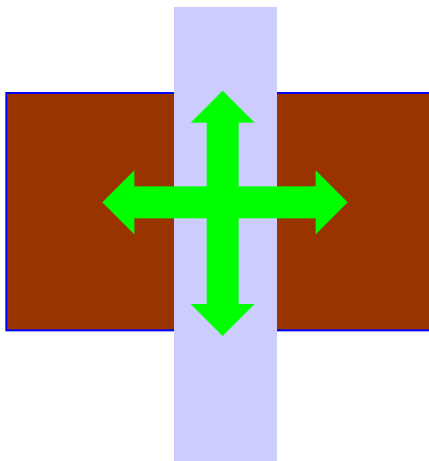
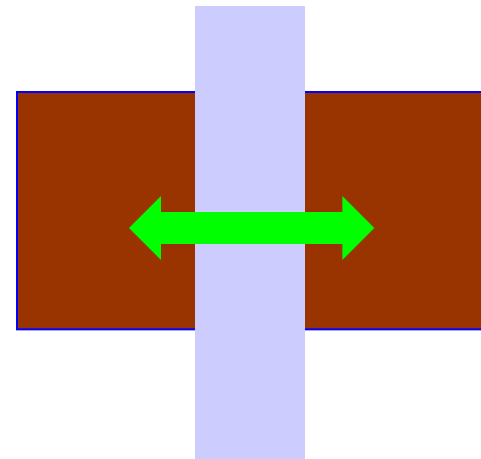


Fig. 3.7. The deformation of a cube under (a) biaxial tensile and (b) [110] uniaxial compressive stress



Biaxial stress



Uniaxial stress

Bandgap as a function of stress

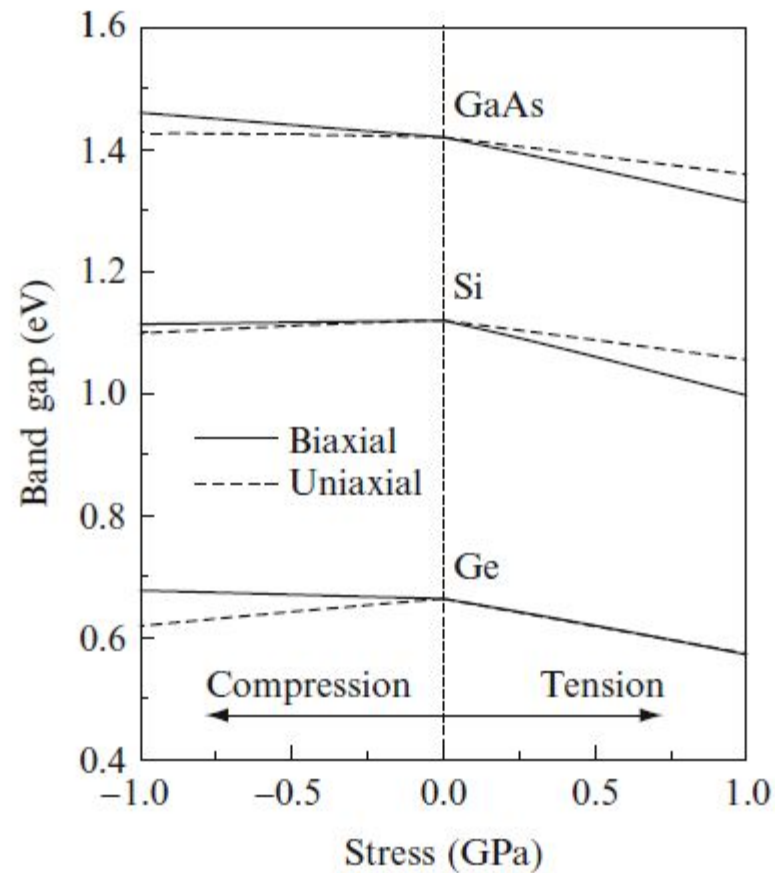


Fig. 4.35. Band gap as a function of stress for GaAs, Si, and Ge

Si bands under stress

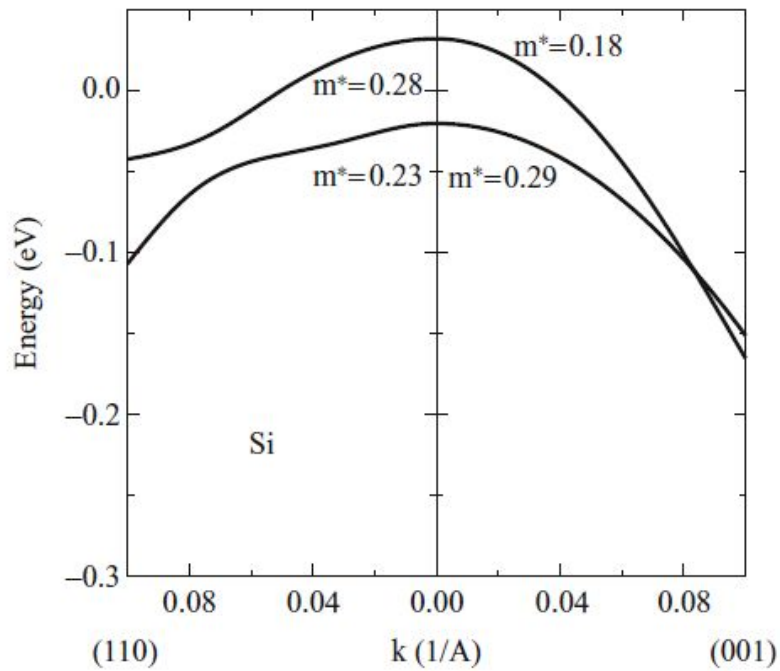
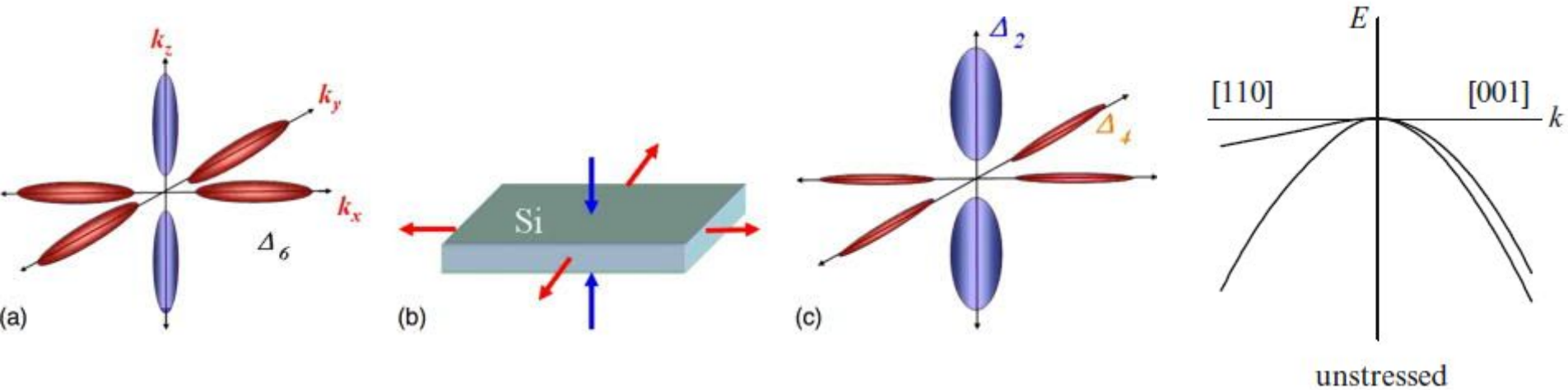


Fig. 4.31. Si valence bands under biaxial tensile stress

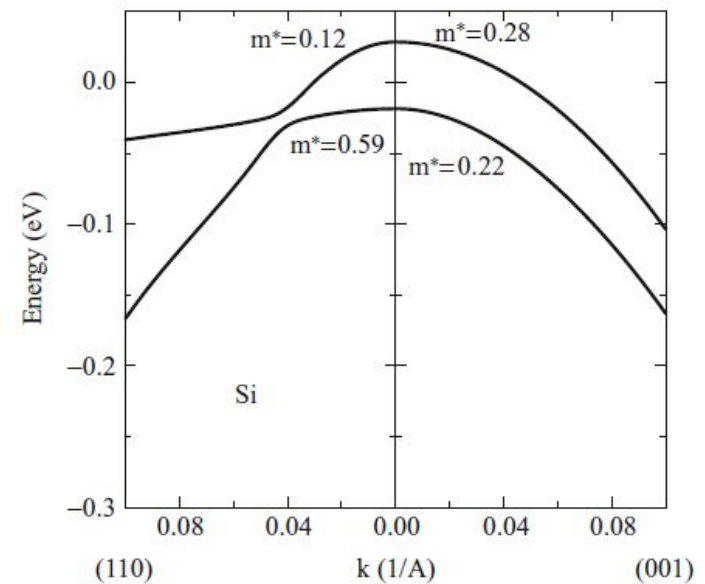
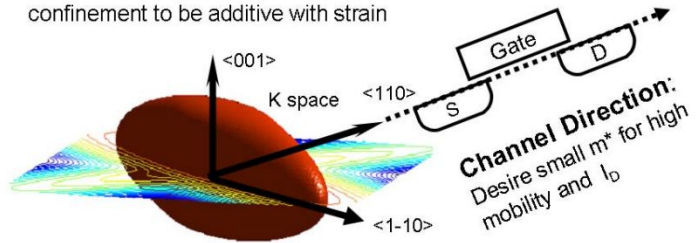


Fig. 4.32. Si valence bands under uniaxial compressive stress

Effective mass wishlist

Out of Plane: Desire large m^* for quantum confinement to be additive with strain



In-plane $\perp$ to Channel:
Desire large m^* for high density-of-states

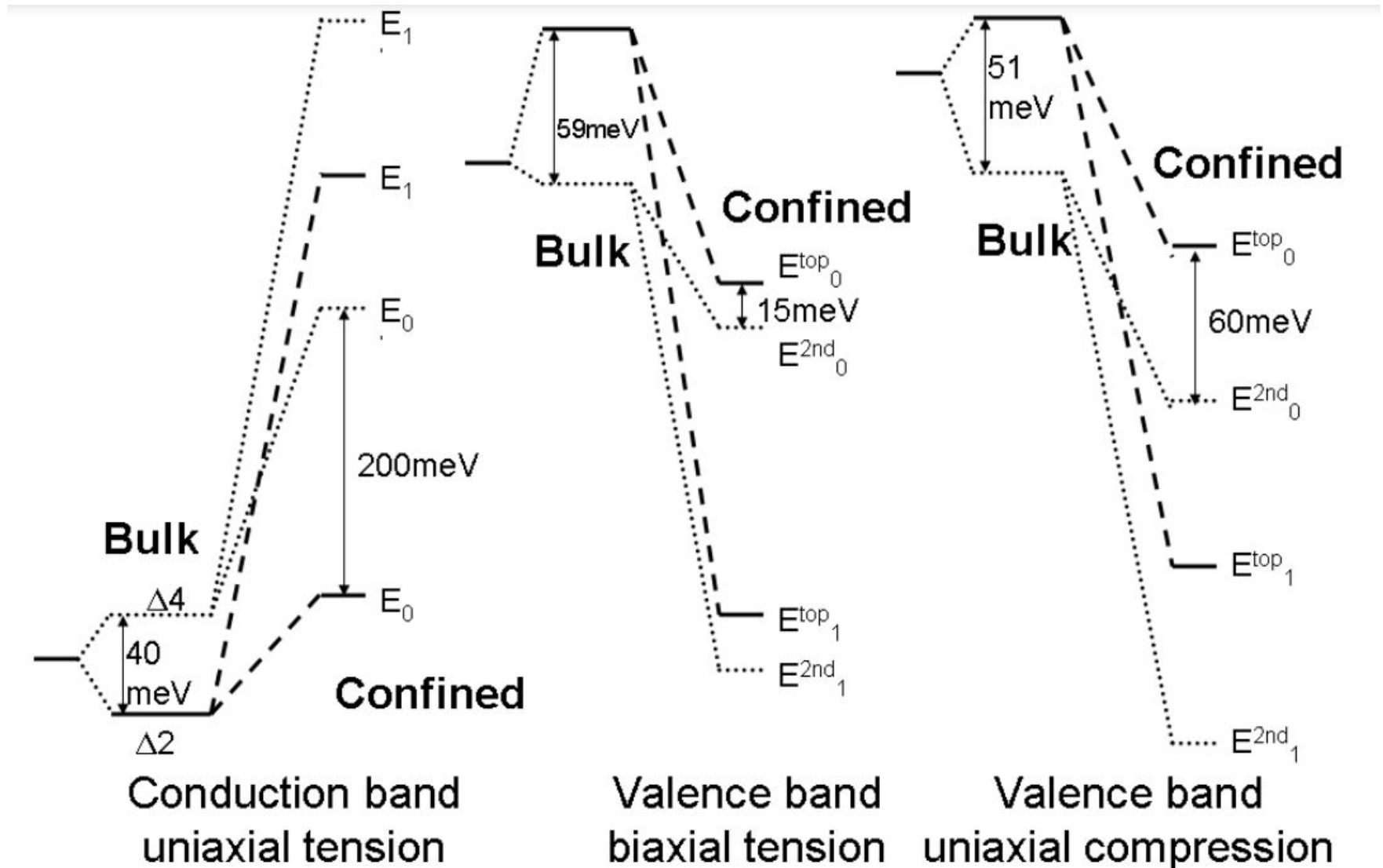
$$\mu \sim \frac{q \langle \tau \rangle}{m_x} \rightarrow \text{want low effective mass}$$

$$m_{\text{DOS}}^{2D^*} = \sqrt{m_x m_y} \rightarrow \text{want high } m_y \text{ for large DOS}$$

$$\Delta E \sim \frac{\hbar^2 k^2}{2m_z} \rightarrow \text{want high } m_z^* \text{ to reduce quantum confinement}$$

Want large subband splitting for maximum occupancy in bottom/top subbands

Effect of stress and quantum confinement



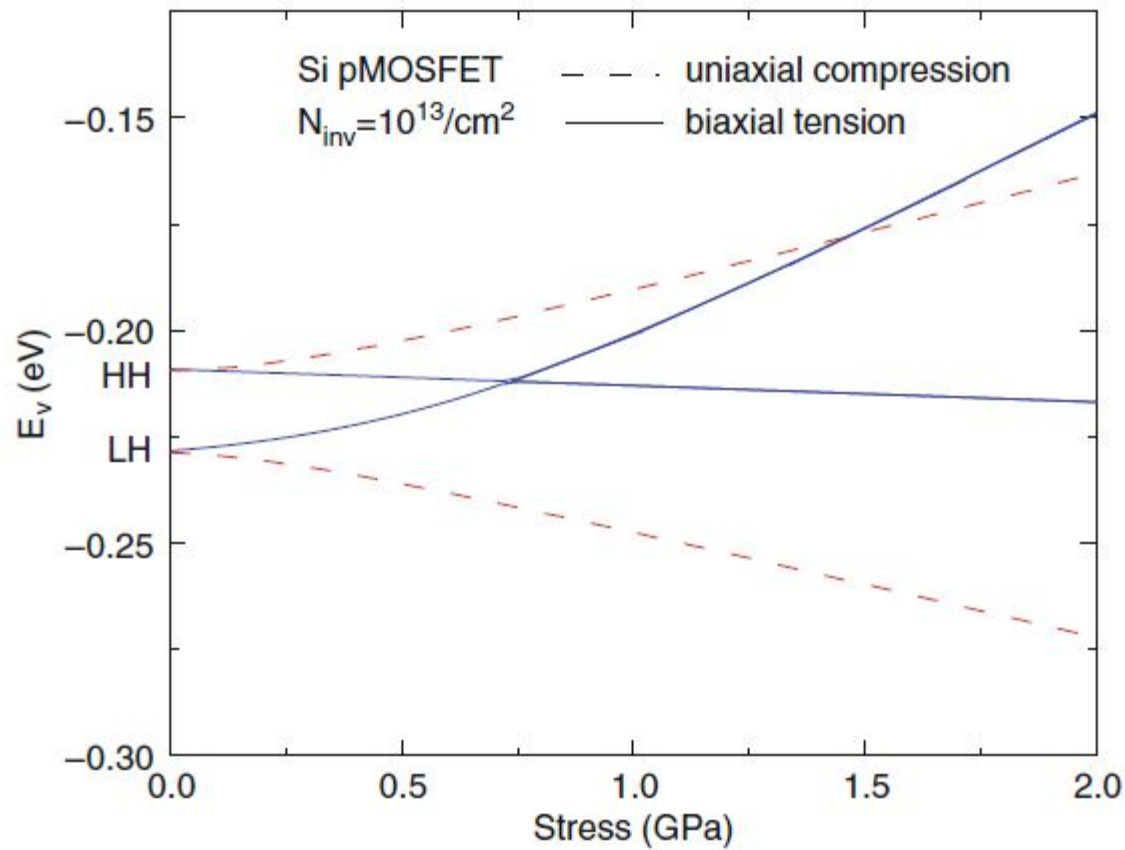
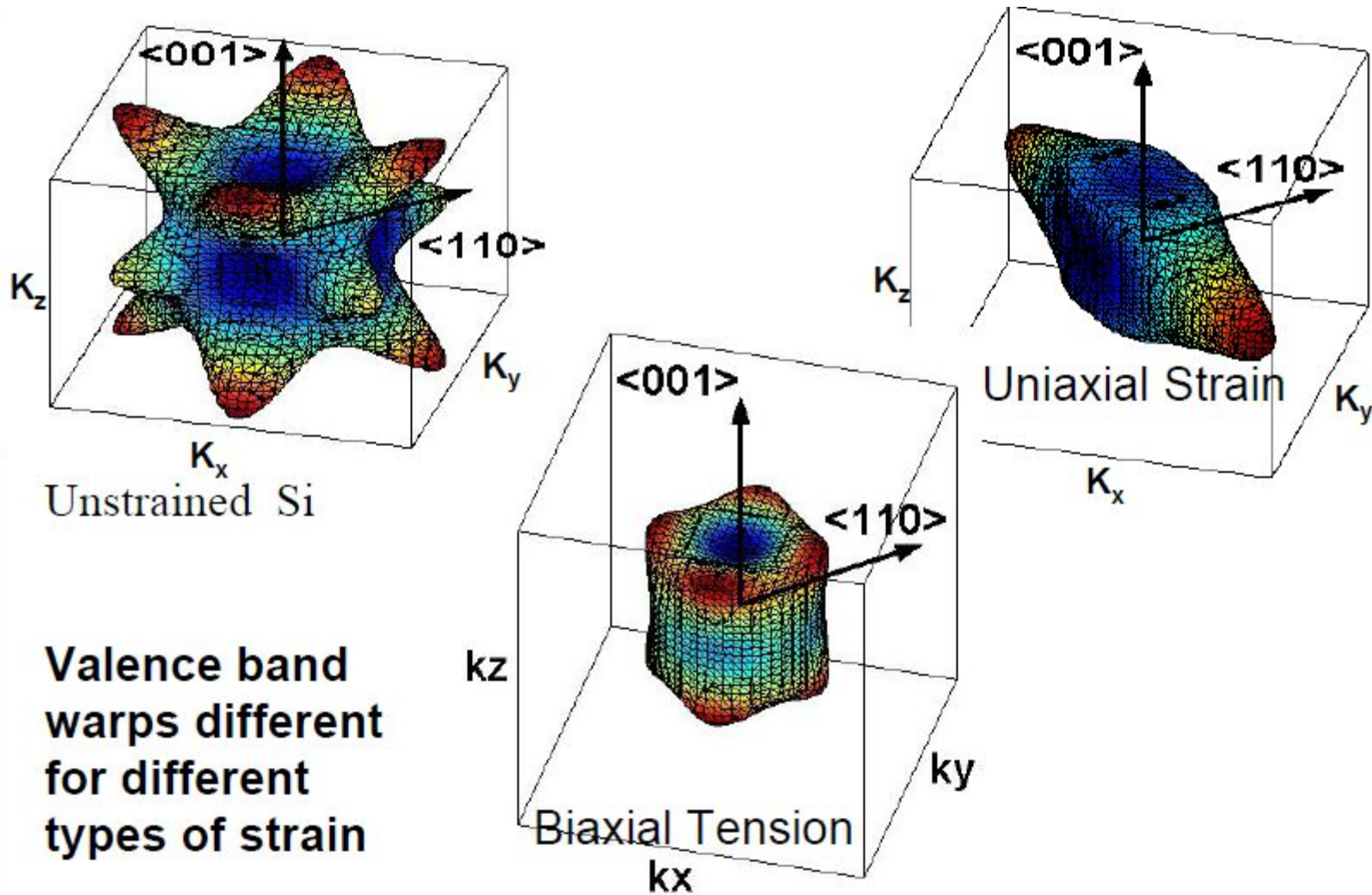


Fig. 5.30. Shift of the ground HH and LH subband for Si 2D hole gas in the Si p-channel MOSFET inversion layer

Valence band warping



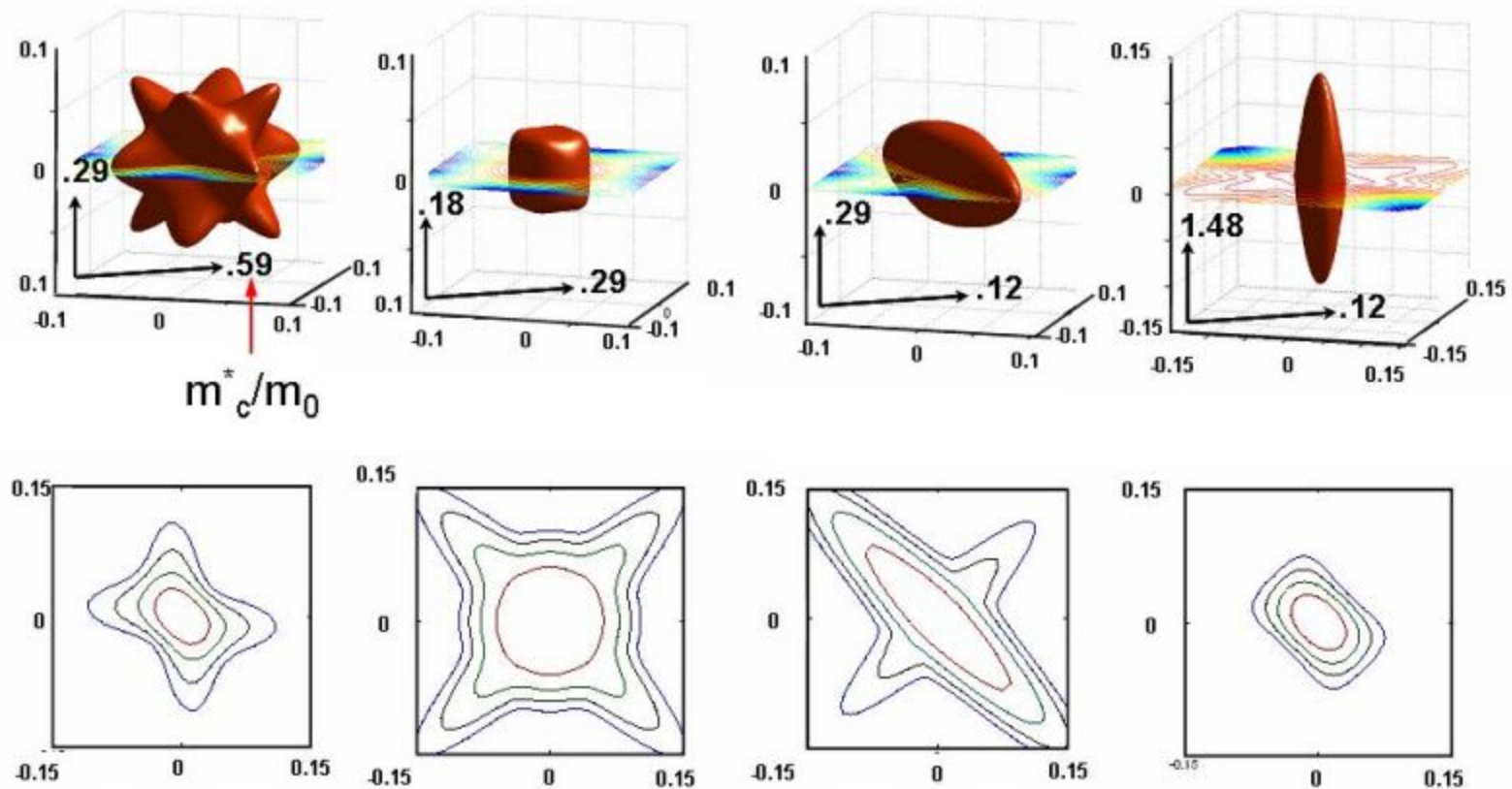


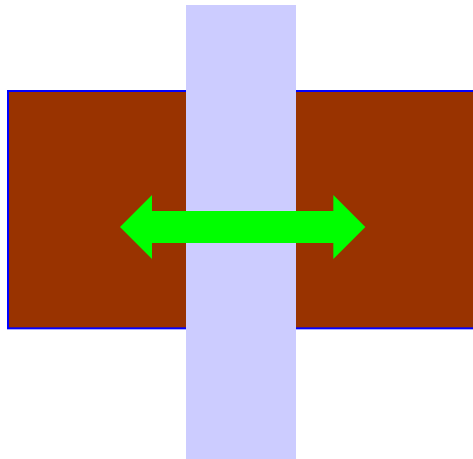
Fig. 7. Constant energy surface and 2D contours under 1 MV/cm confinement in k space for top valence band for: unstressed Si; 1 GPa biaxial tensile stress; 1G Pa uniaxial compressive stress on (001) Si; 1GPa uniaxial compressive stress on (110) Si.

How is mechanical stress introduced ?

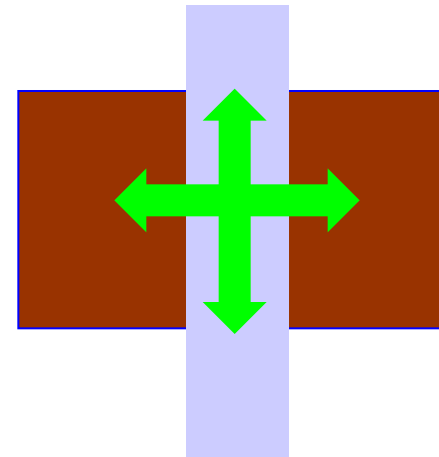
- Due to material mismatch of A and B from material composition, element size/volume and thermal expansion rate, there will be a stress transfer from one material to another
- Stress transfer depends on
 - a) material growth or deposition conditions
 - b) subsequent anneal or implantation steps

Material A (tens/comp)
(eg. Nitride)

Material B
(eg. Si)



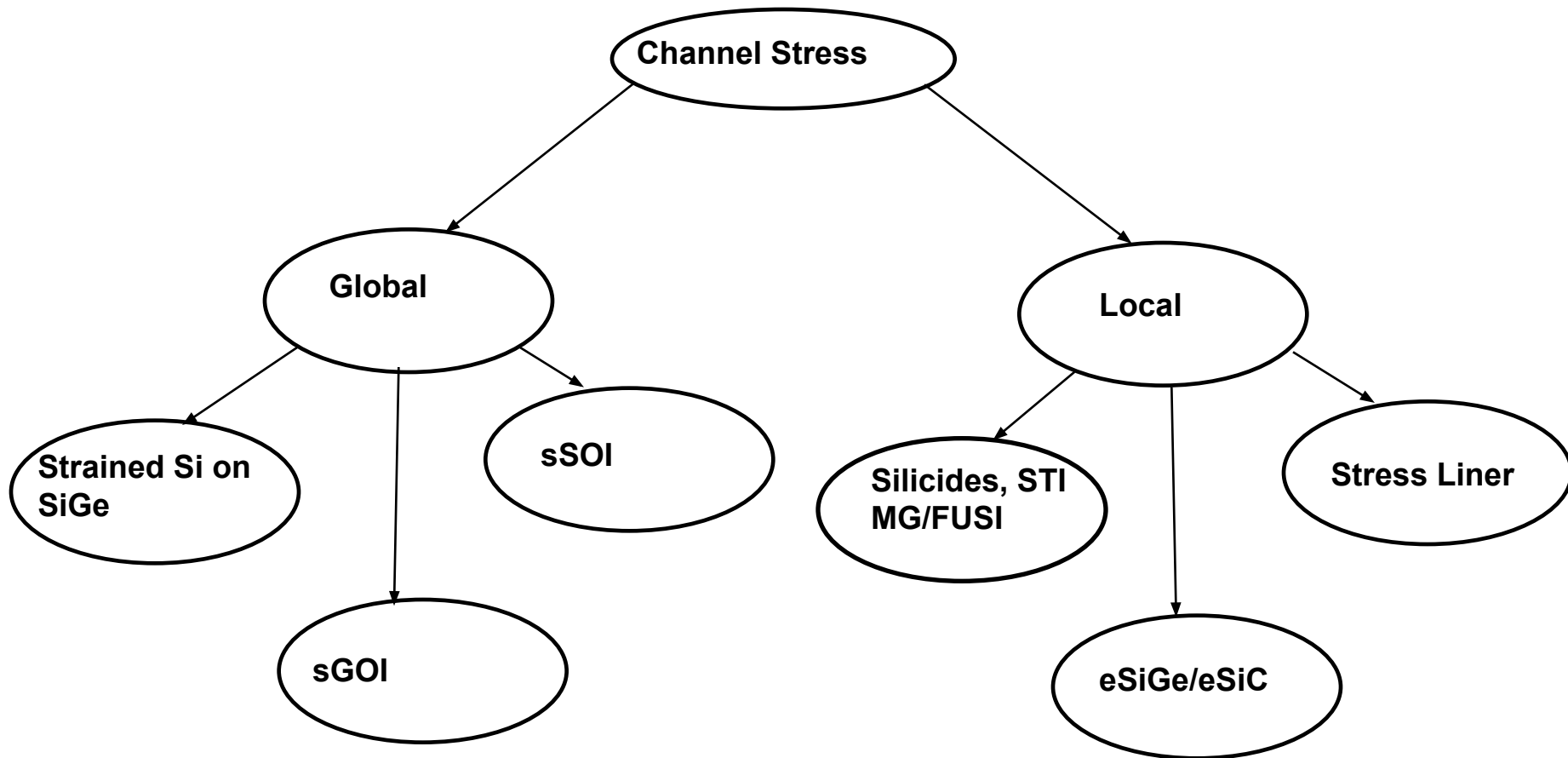
Uniaxial stress



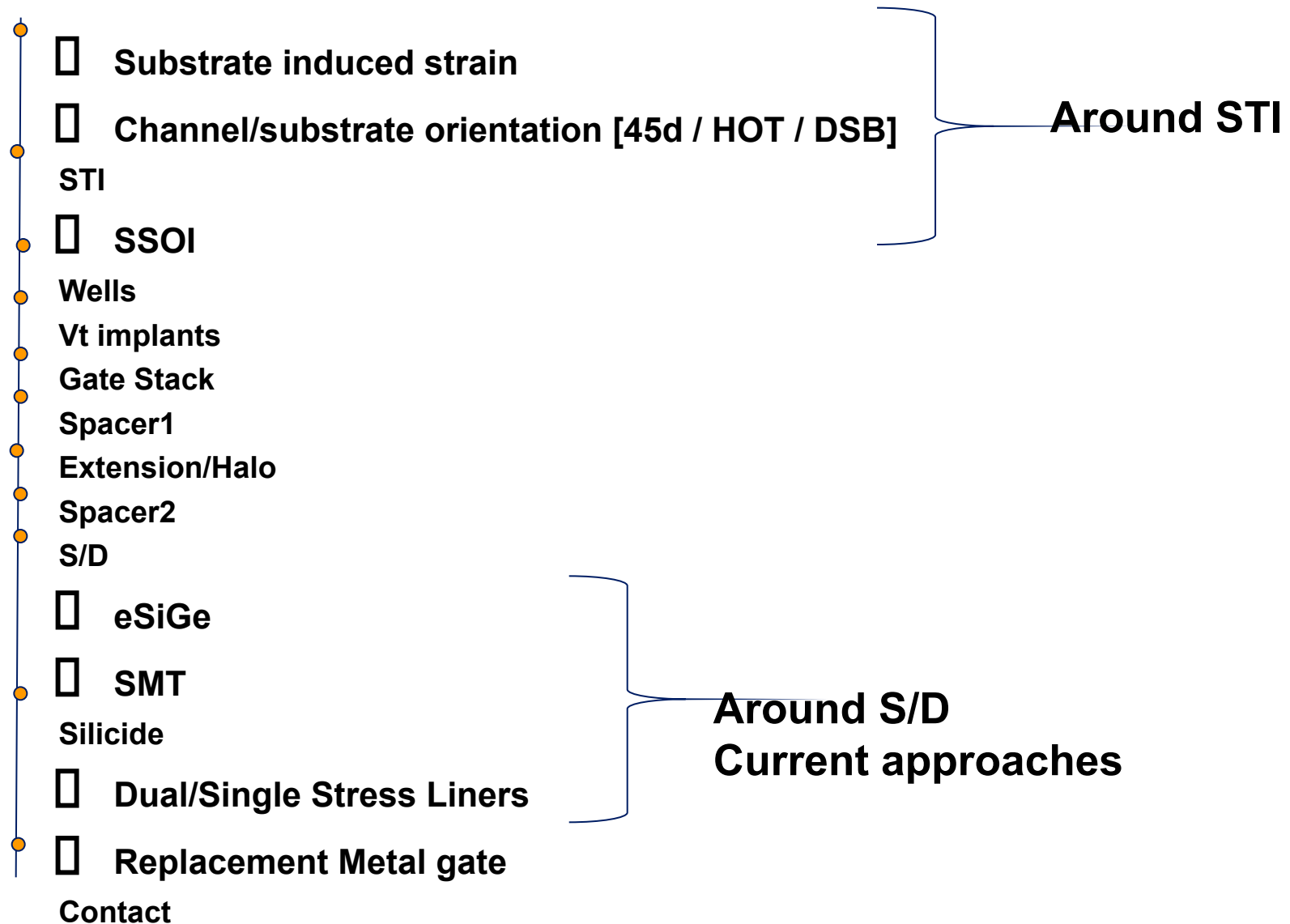
Biaxial stress

Where can we bring stress in our devices ?

34



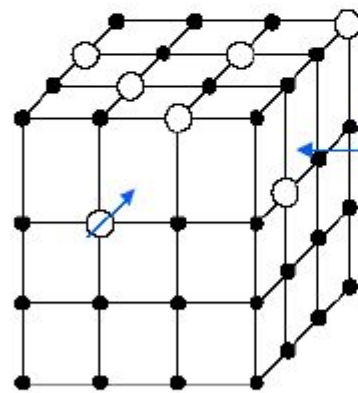
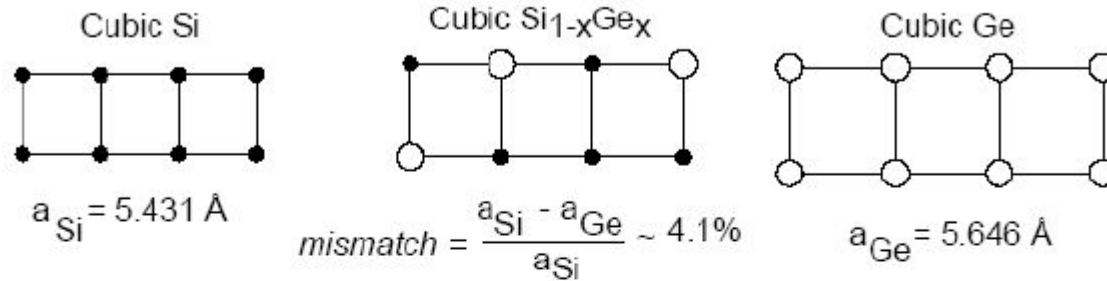
Where they appear ?



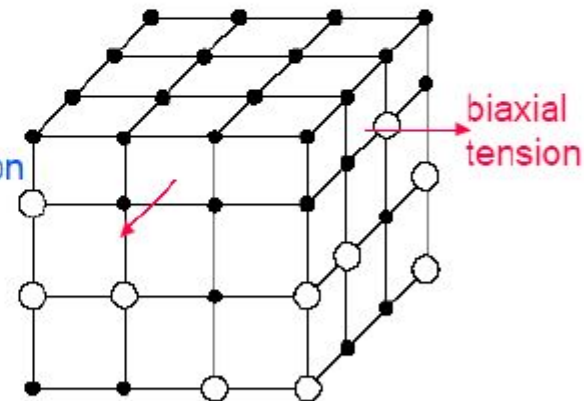
Substrate induced strain

Si and SiGe crystal structures

Equilibrium (relaxed) Lattices



Strained $\text{Si}_{1-x}\text{Ge}_x$ on Relaxed Si
(heterojunction bipolar transistors)

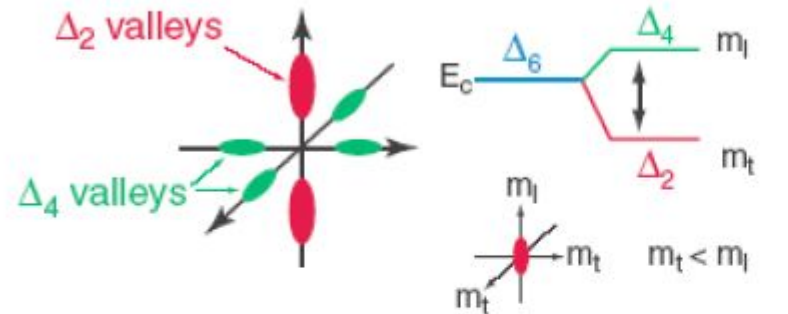


Strained Si on Relaxed $\text{Si}_{1-x}\text{Ge}_x$
(field-effect transistors)

Source: J. Hoyt, MIT

Substrate induced strain (bi-axial tension)

Conduction band

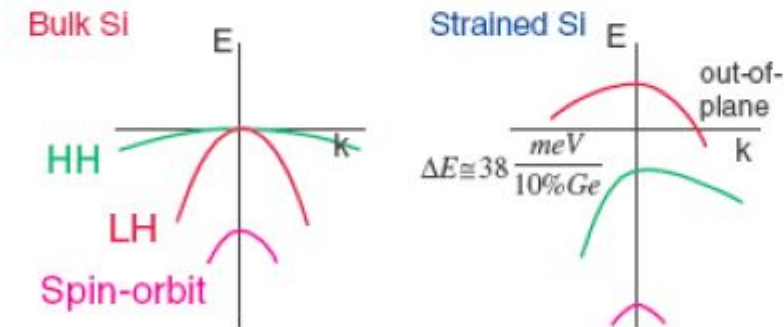


Conduction Band

Additional splitting: →
Band repopulation

- reduced intervalley scattering
- smaller in-plane effective transport mass

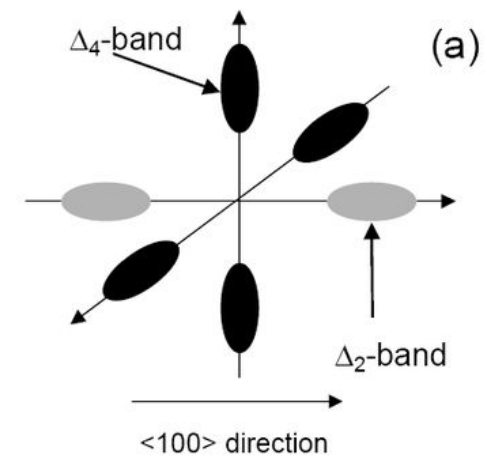
Valence band



Valence Band

HH/LH degeneracy lifted at Γ

- reduced interband scattering
- smaller in-plane transport mass due to band deformation



Δ_2 -band:

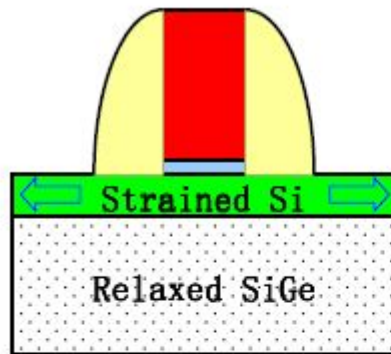
$$m_{\perp} = m_l = 0.916m_0, \quad m_{\parallel} = m_t = 0.196m_0$$

Δ_4 -band:

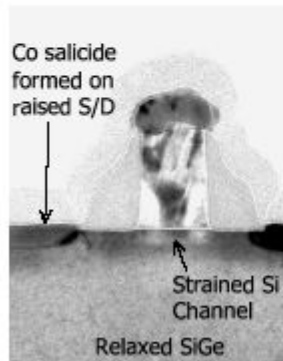
$$m_{\perp} = m_t = 0.196m_0, \quad m_{\parallel} = (m_l m_t)^{1/2} = 0.42m_0$$

Source: J. Hoyt, MIT

Substrate induced strain (bi-axial tension)

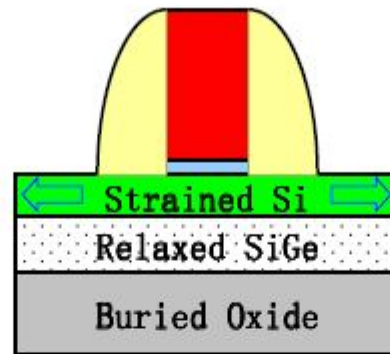


Strained Si/SiGe
Bulk MOSFET

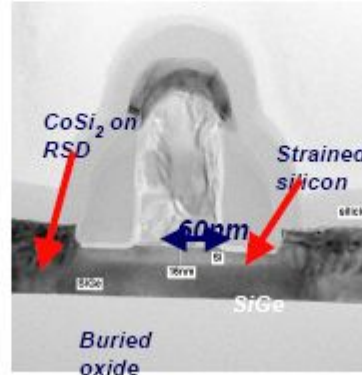


K. Rim et al., *Symp. VLSI Tech.*, p. 59, 2001.

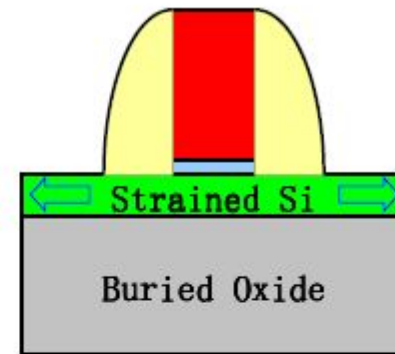
K. Rim et al., *Symp. VLSI Tech.*, p. 98, 2002.



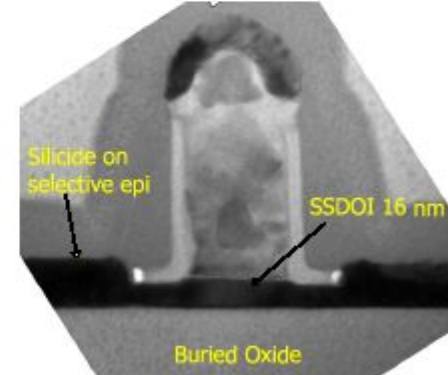
SGOI (SiGe-on-
Insulator) MOSFET



B. Lee et al., *IEDM* 2002

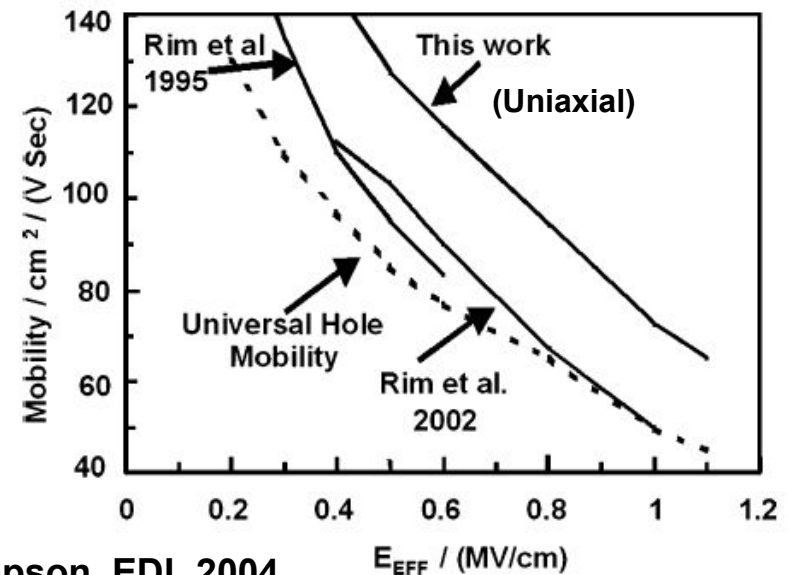
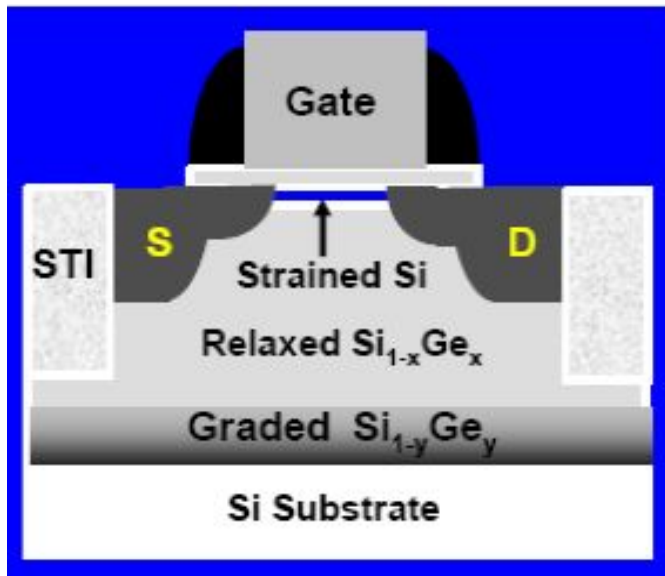


SSDOI MOSFET



K. Rim et al., *IEDM*, 2003.

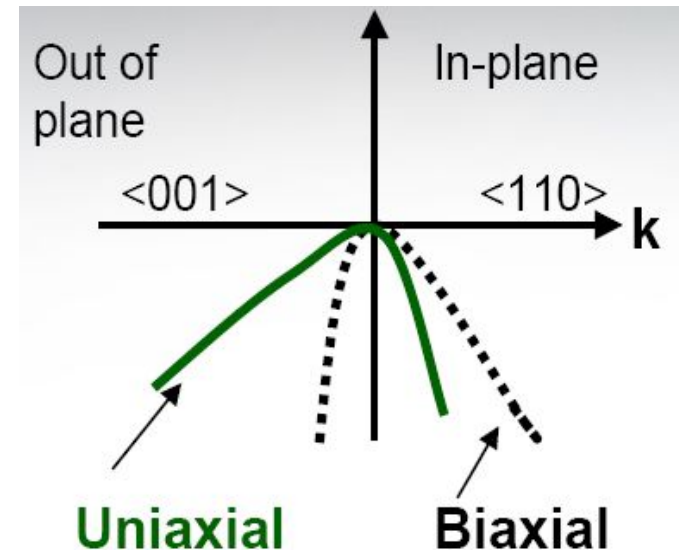
Why not bi-axial tension ?



S.E. Thompson, EDL 2004

Despite studied for many years, biaxial stress was not adopted:

- ❑ Integration difficulties (dislocations, Ge up diffusion, cost)
- ❑ Poor hole mobility gain at high E_{eff}
- ❑ Large strain induced v_t shift for NMOS
- ❑ At high E_{eff} , subband separation is small

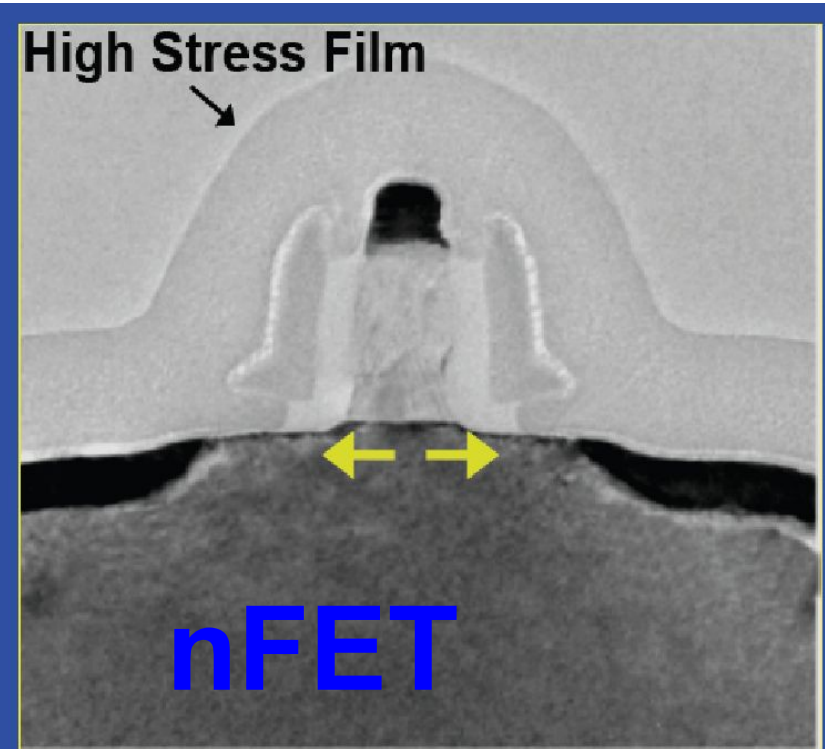
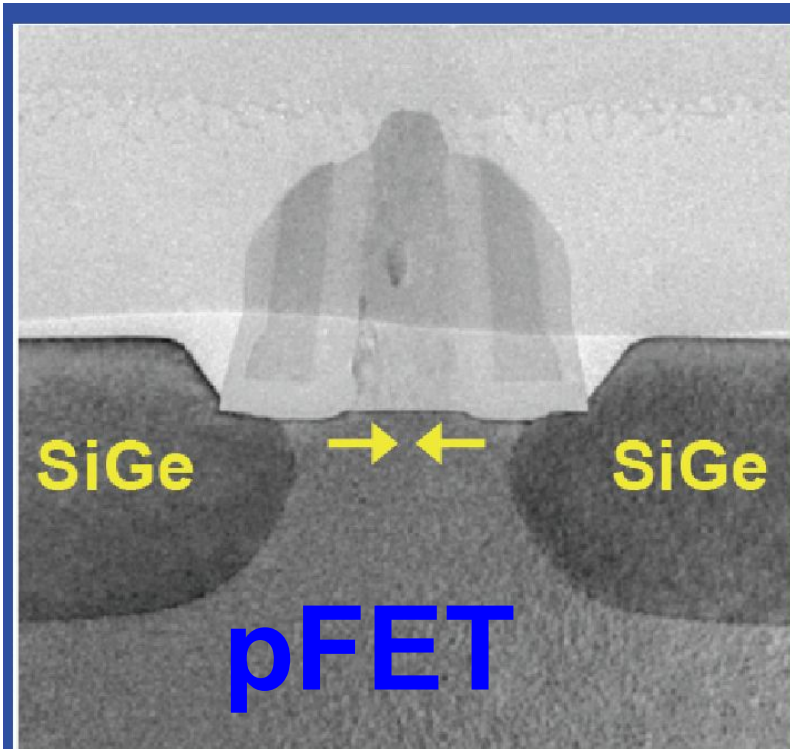


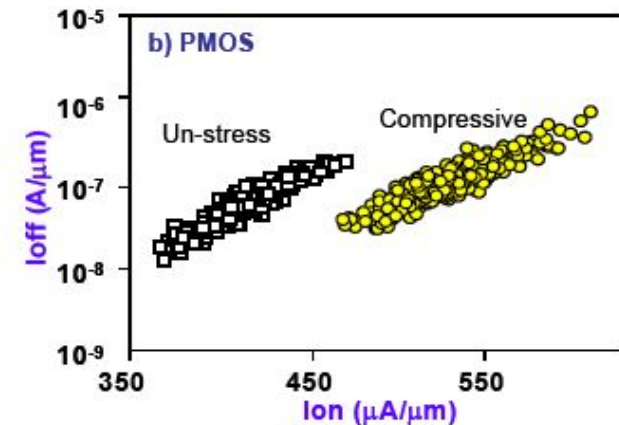
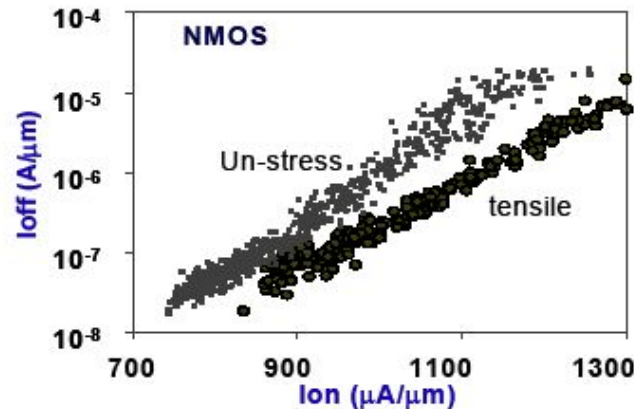
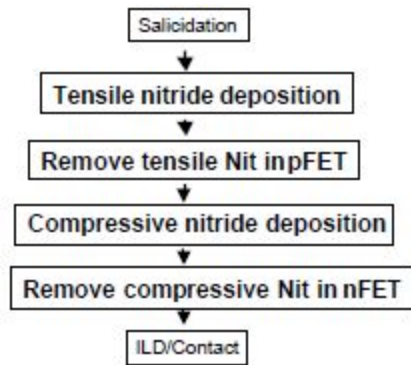
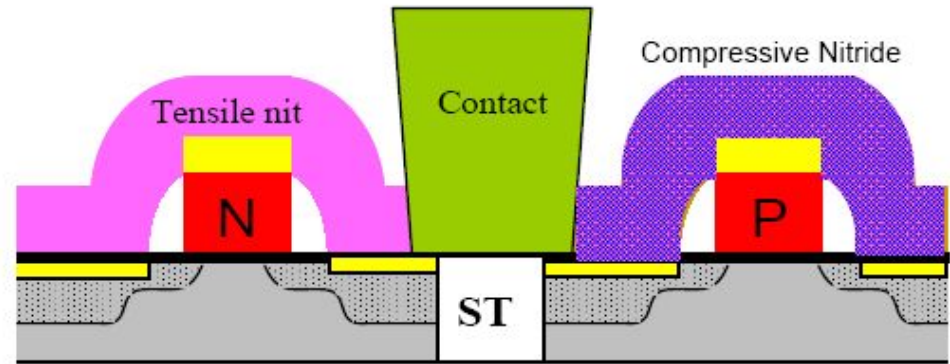
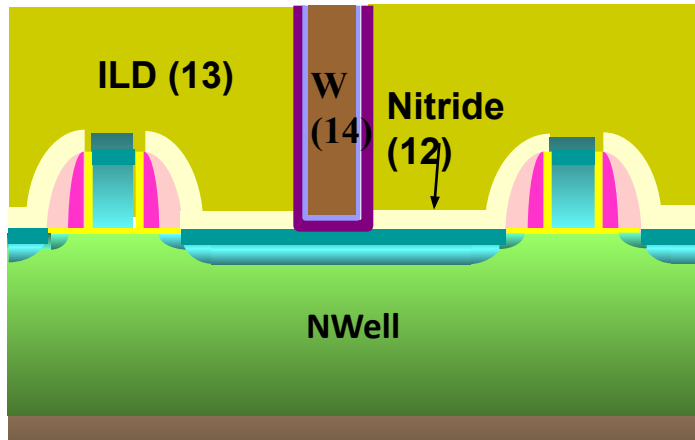
S.E. Thompson, IEDM 2004

Intel 90nm

- Compressive uniaxial stress in direction current flow increases hole mobility.
 - 50% - 70% gain in drive current.
- Tensile uniaxial stress in direction of current flow increase electron mobility.
 - 10% - 15% gain in drive current.

K. Mistry et al, VLSI Tech. Sym. 2004

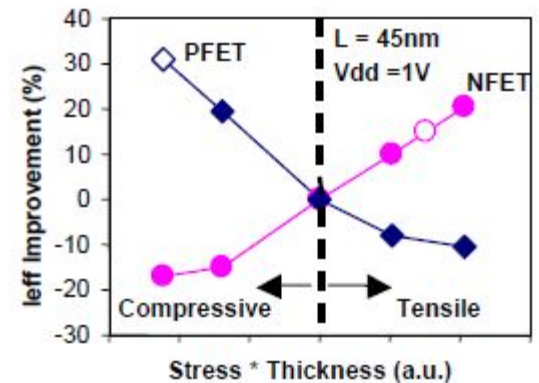




Dual Stress Liner-

Tensile liner for NFET

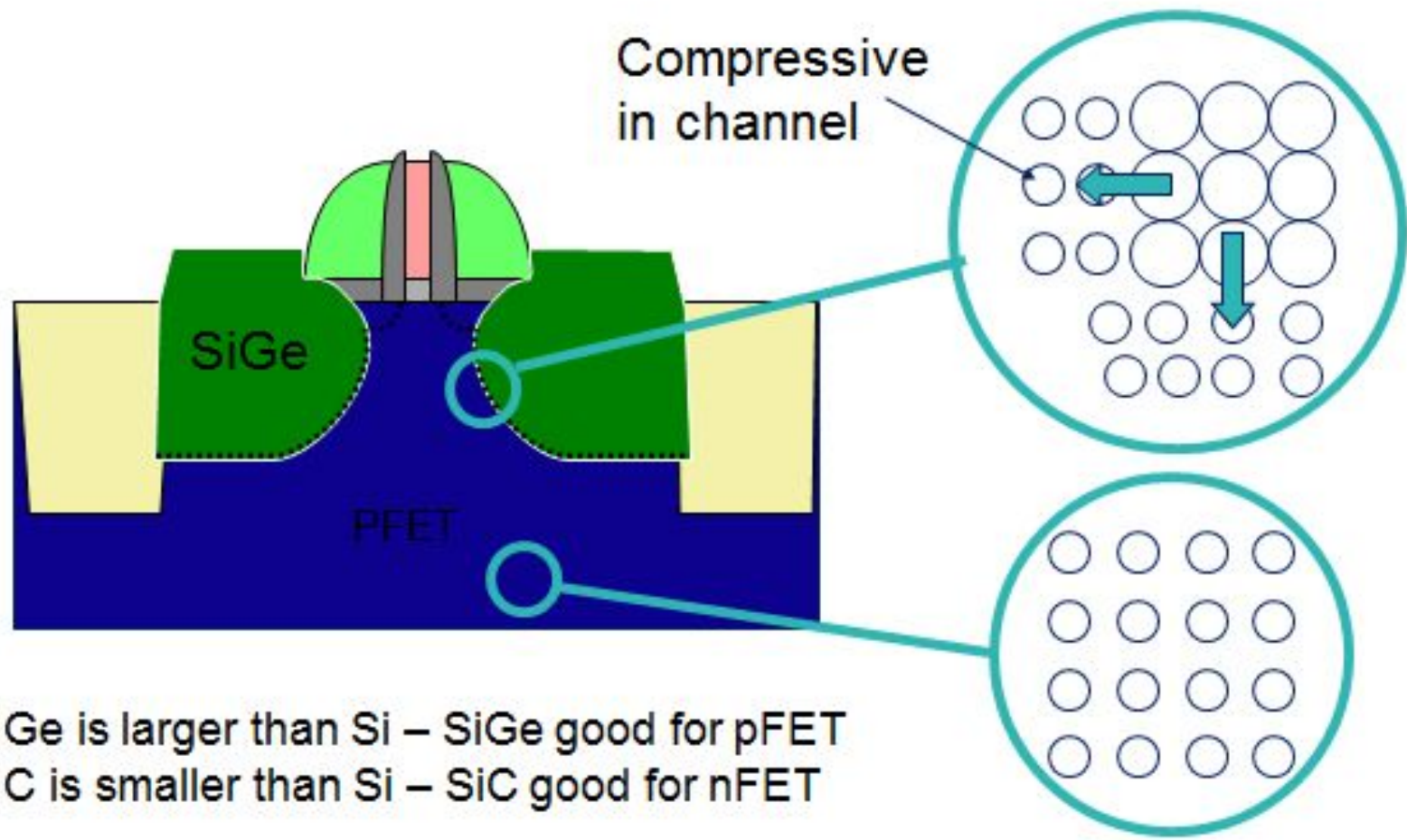
Compressive liner for PFET



Stress liner

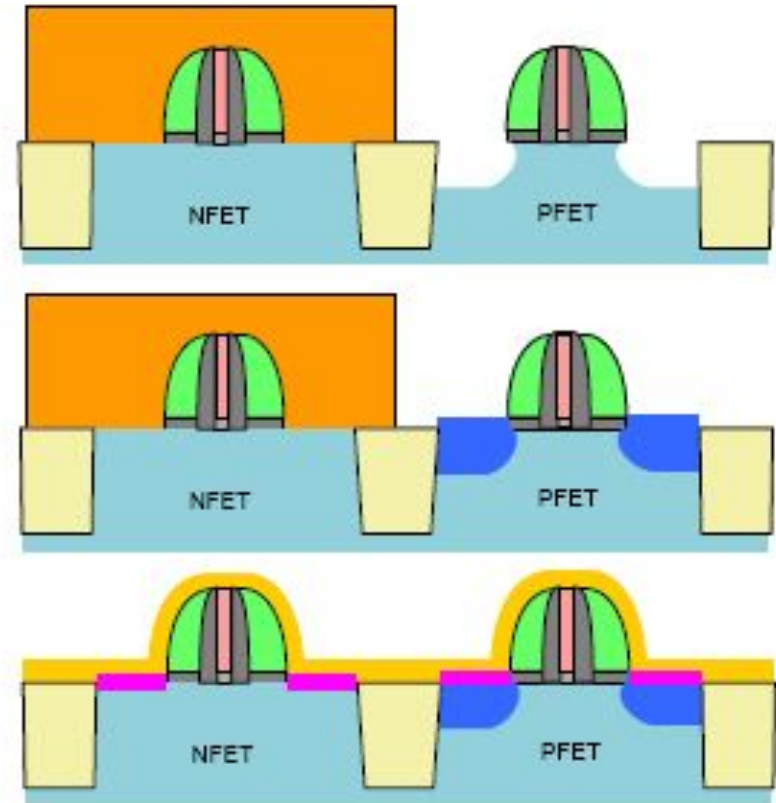
- **SiN films deposited using ammonia, silane and nitrogen**
- **Deposition method – PECVD 300-500C
(Silicide sets limit)**
- **Tensile films contains lower bonded hydrogen while compressive films contain more**
- **Film could be single or multilayered**

Embedded SiGe (eSiGe)

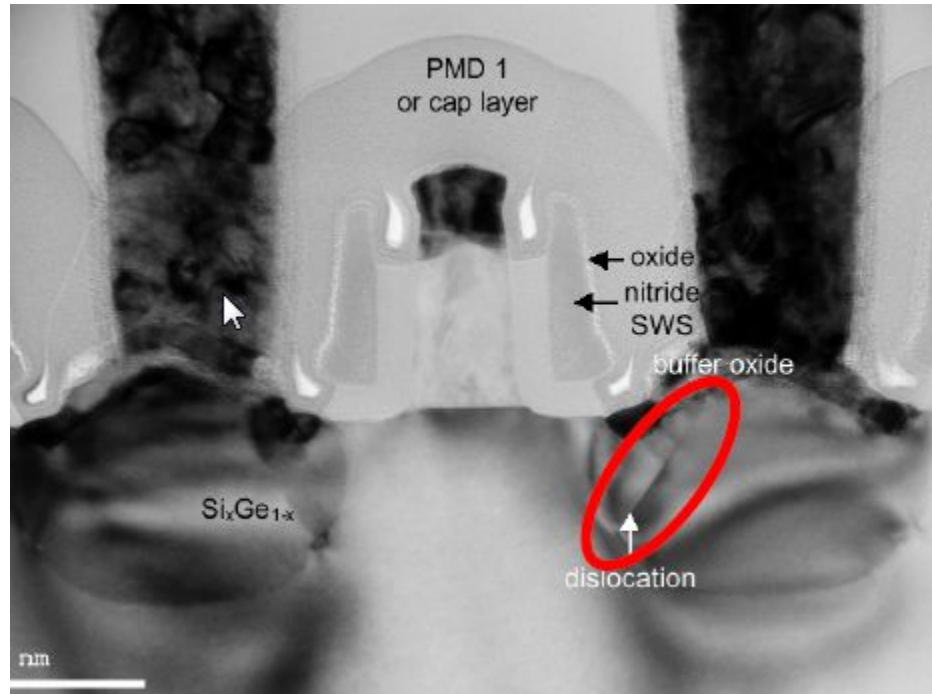


eSiGe Process Description

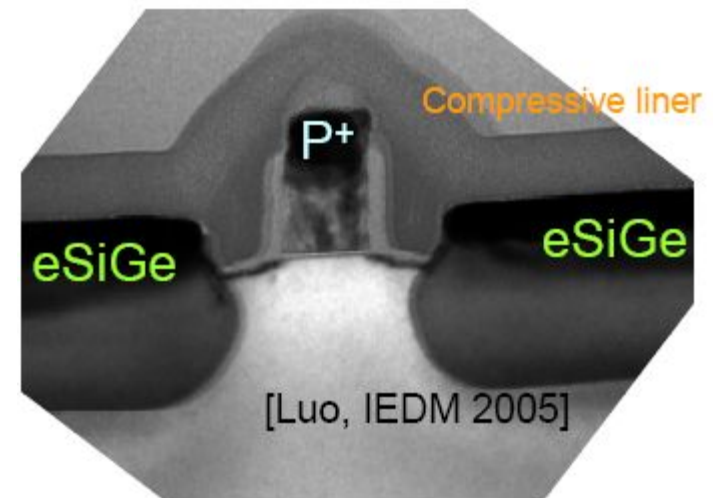
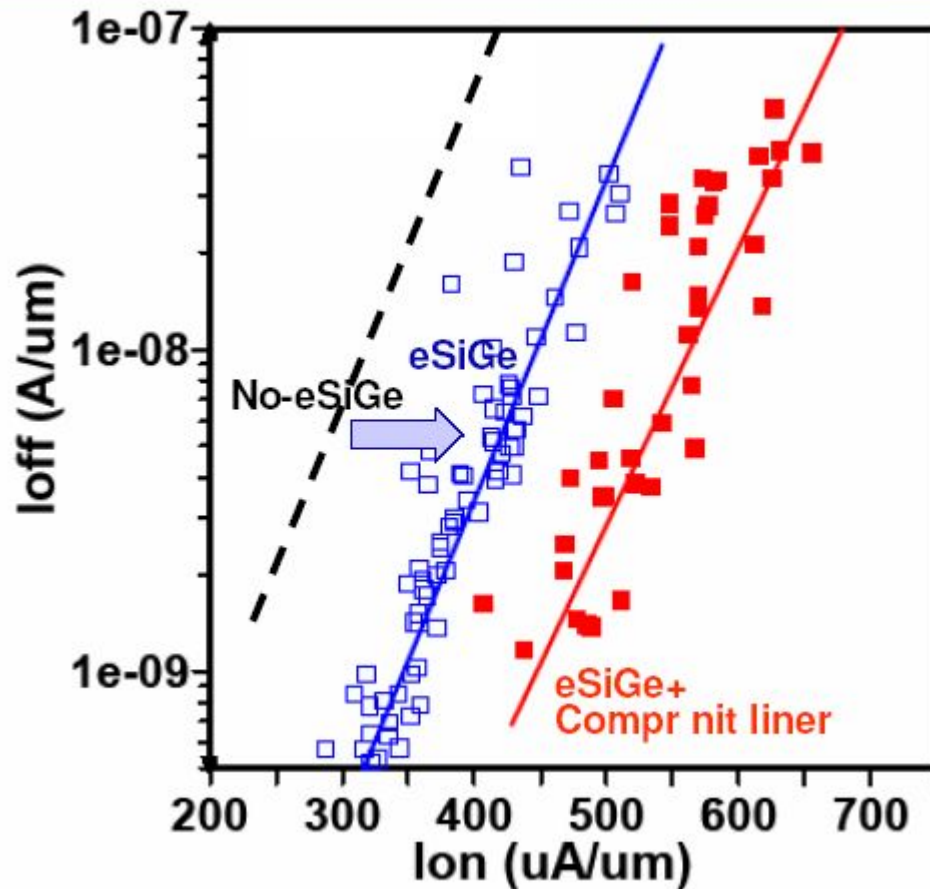
1. Form protective layer over NFET
2. RIE to recess Si
3. Selectively deposit eSiGe
4. Remove protective layer
5. Ion implant and anneal source/drain
6. Form Ni silicide(-germanide)



Process integration challenges - dislocations



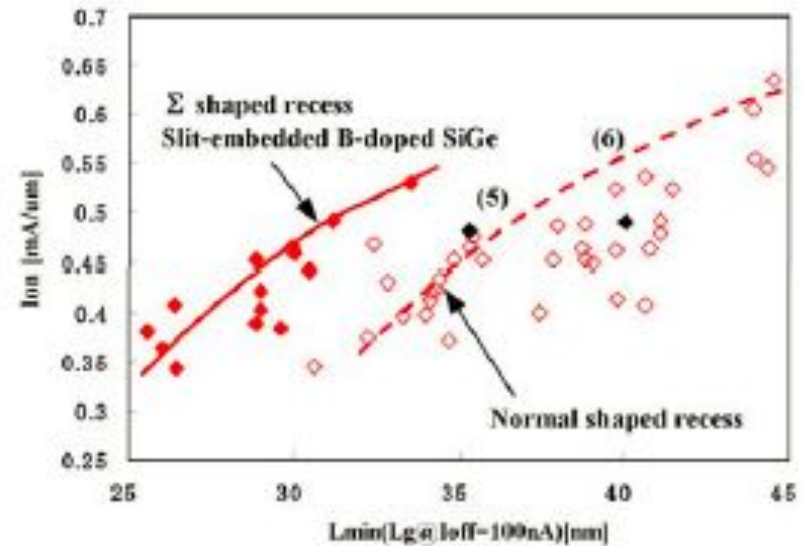
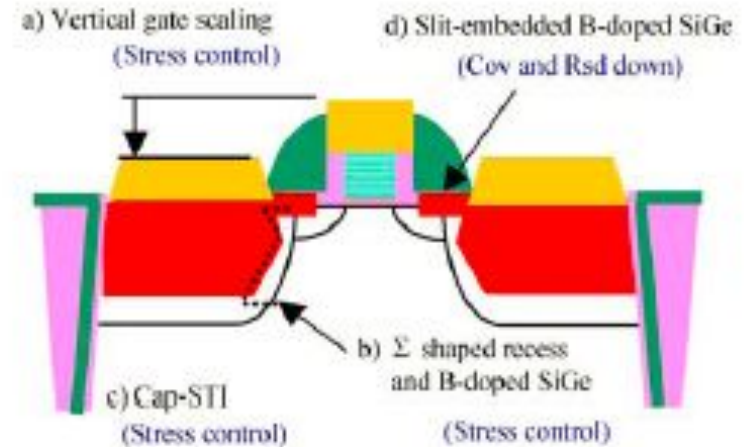
Can we add the benefits of eSiGe and stress liner ?



Embedded SiGe Cross-section

SiGe profile engineering

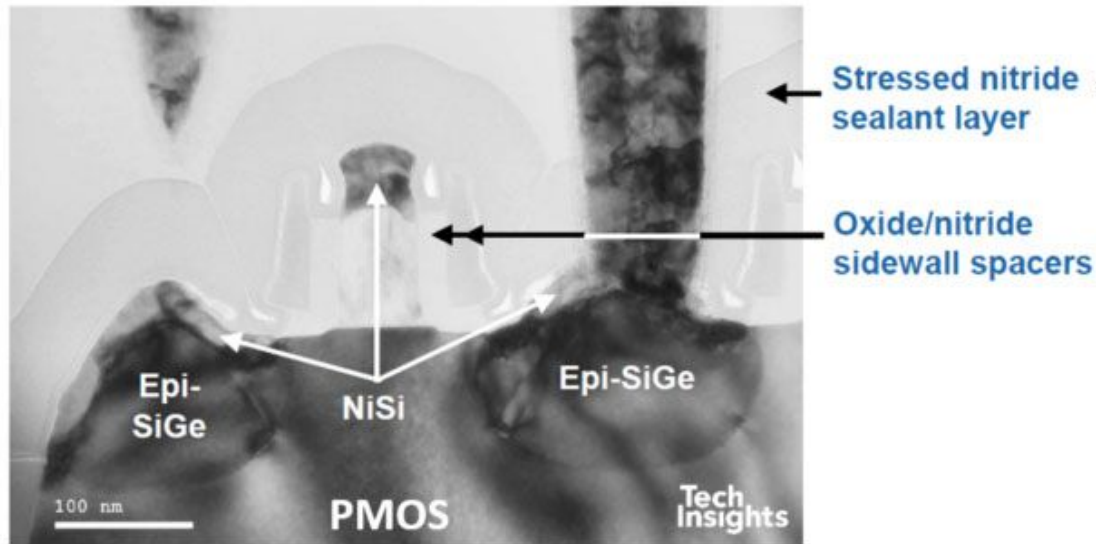
Benefits decrease with generation due to pitch scaling unless new techniques can increase stress in film (improve gap fill, better eSiGe profile, increasing Ge %)



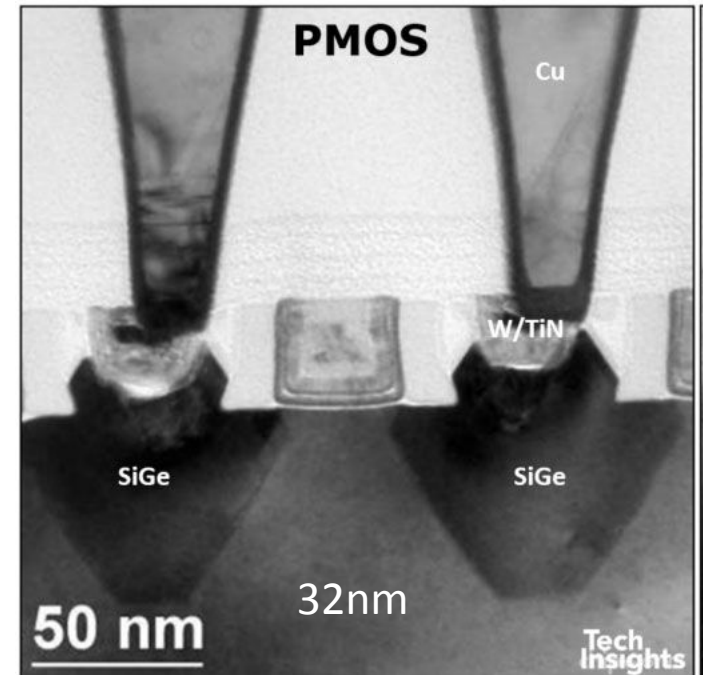
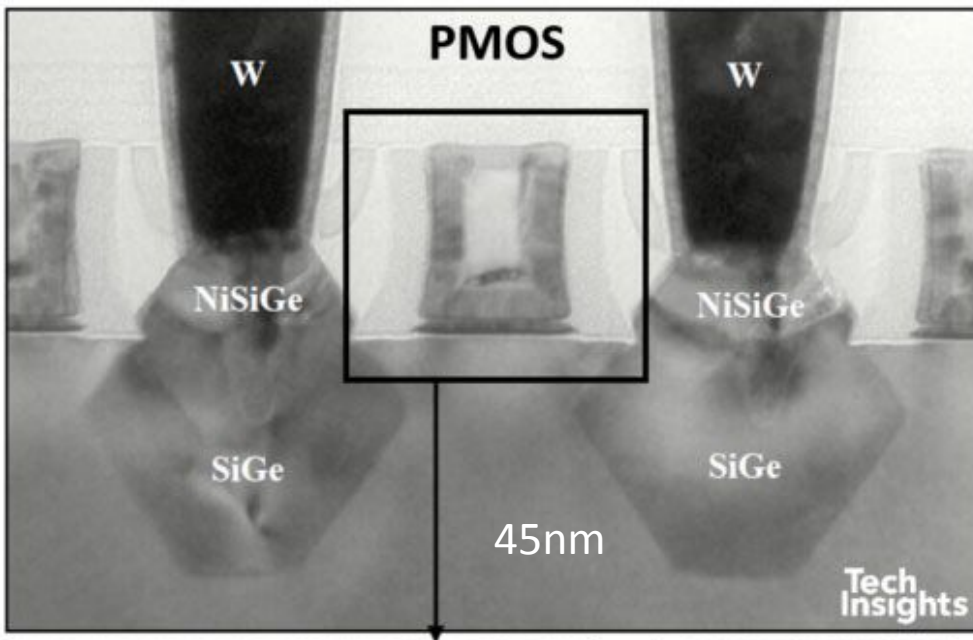
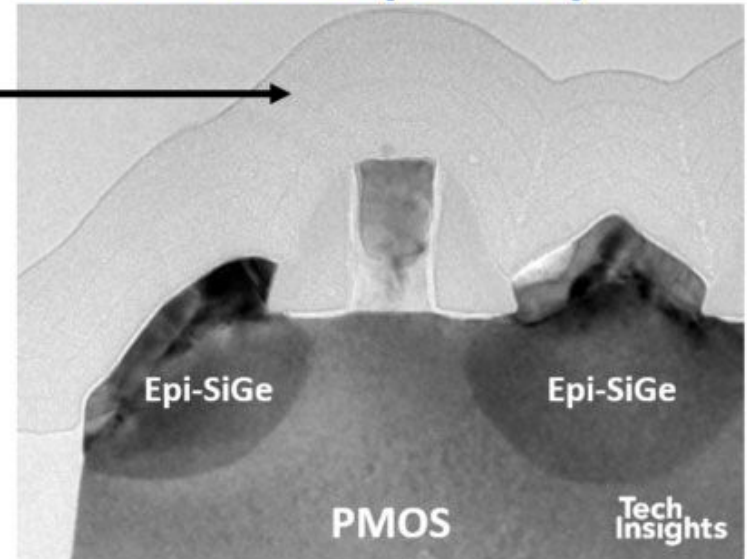
SiGe profile engineering

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Intel Prescott (90 nm)



Intel Presler (65 nm)



Embedded silicon carbon for NFET

- Similar idea as e-SiGe for PFET
- Si-C alloy ($\text{Si}_{1-y}\text{C}_y$, Si:C) has a lattice constant smaller than Si if the C atoms stay at substitutional sites of Si crystal
- Embedded Si:C (e-Si:C) in S/D of NFET generates tensile stress in channel to enhance the electron mobility
- Lattice change caused by each percent of substitutional C is 10 times of the lattice change caused by each percent of Ge in Si
- Straightforward idea but really difficult to implement

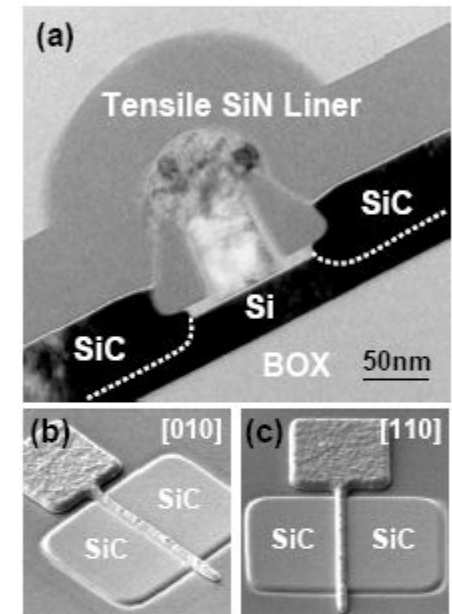
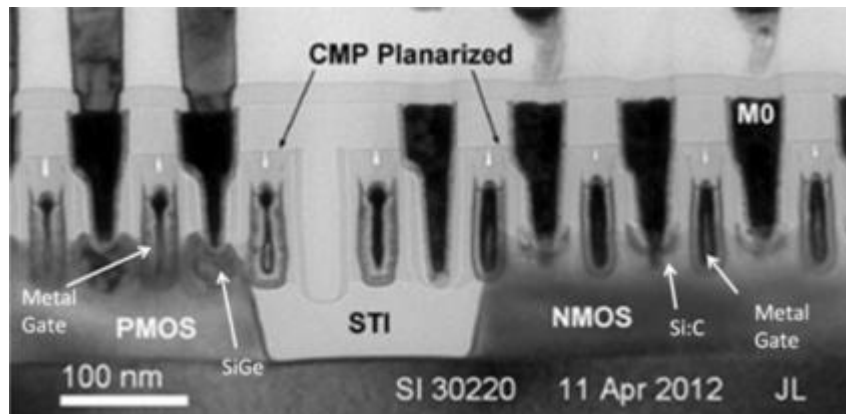
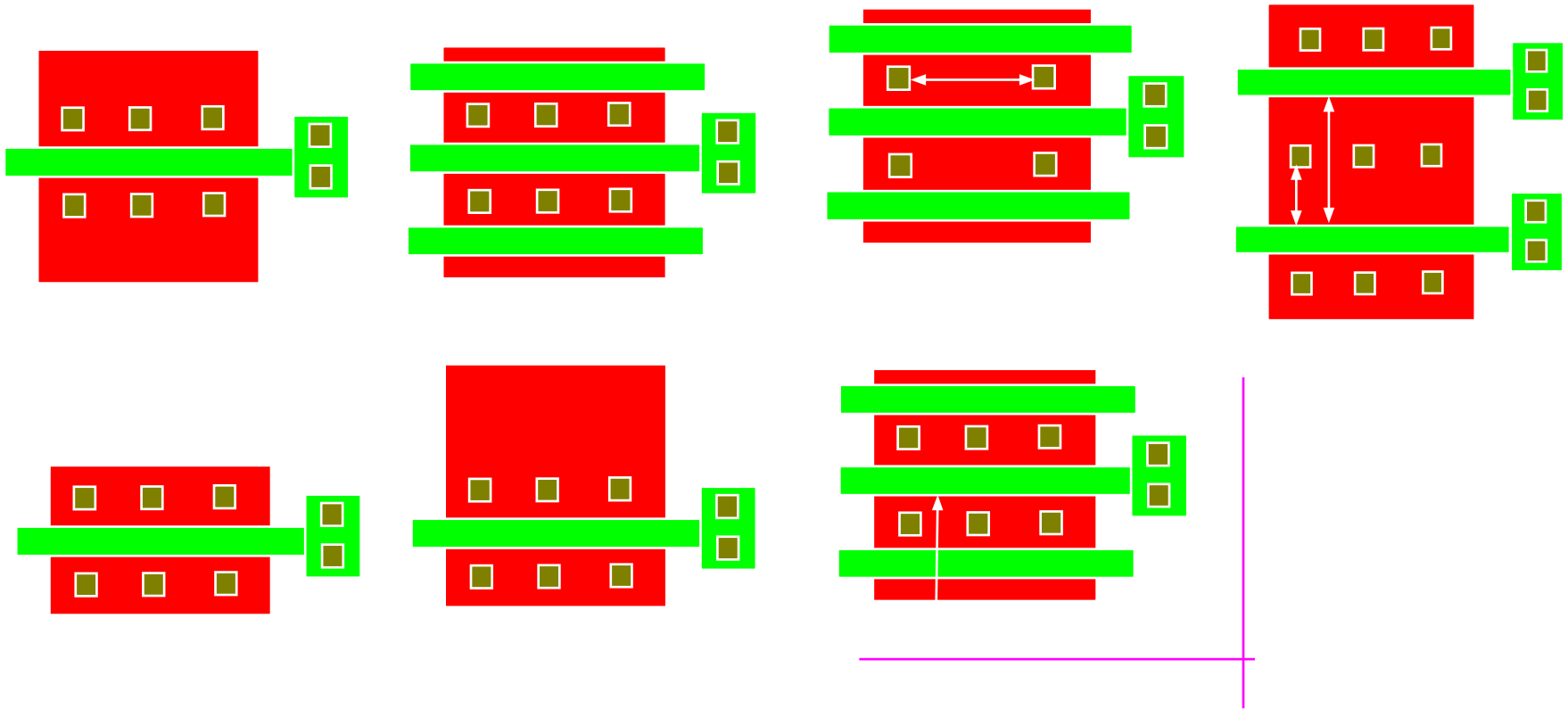


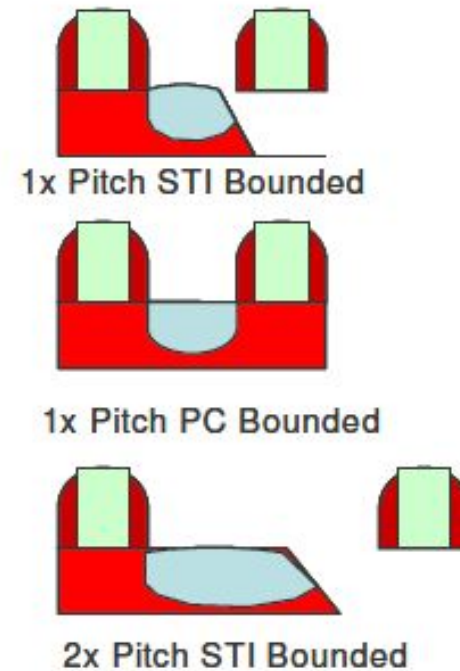
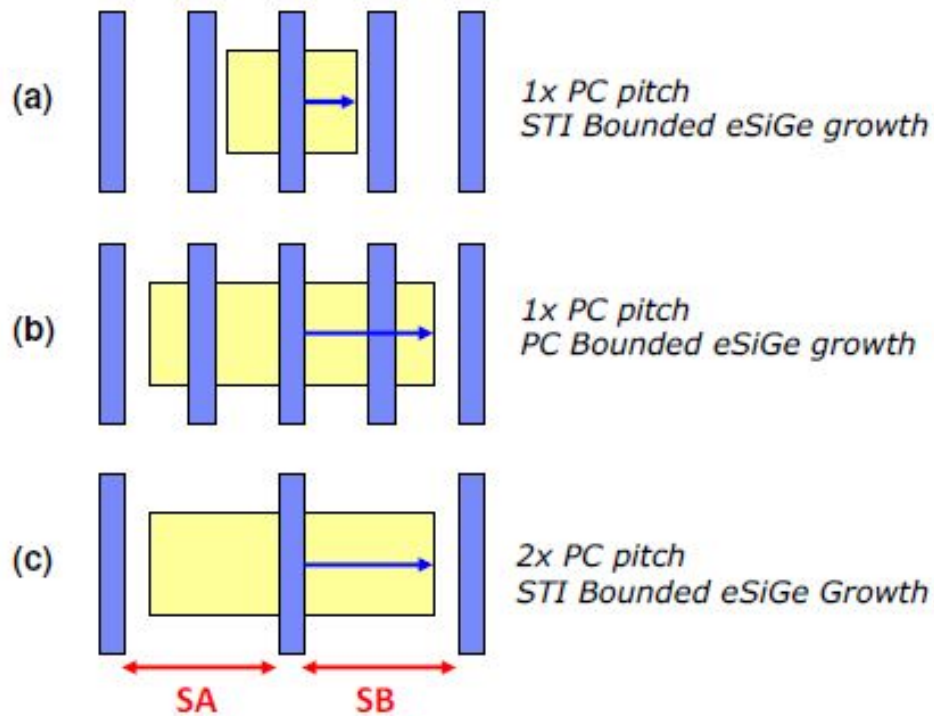
Fig. 1. (a) TEM image of a SOI n-MOSFET with SiC S/D and tensile stress silicon nitride liner. (b) and (c) show transistors oriented along [010] and [110] channel directions, respectively.

Problem with Stress elements



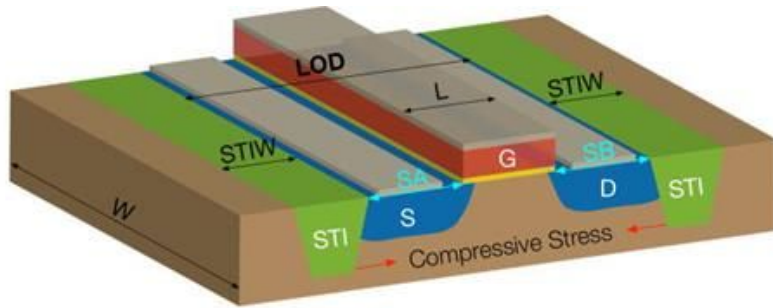
- **Layout dependency** - Stress introduced by liner films or epi S/D is proportional to volume and its proximity to channel
- **Benefits decrease with generation** due to pitch scaling unless new techniques can increase stress in film (improve gap fill, better eSiGe profile, increasing Ge %)
- **Reduction in gate height** also reduces stress efficiency
- **Reliability issues** like plasma charging on thick oxide FETs
- **Dislocations**

Example of eSiGe Layouts:

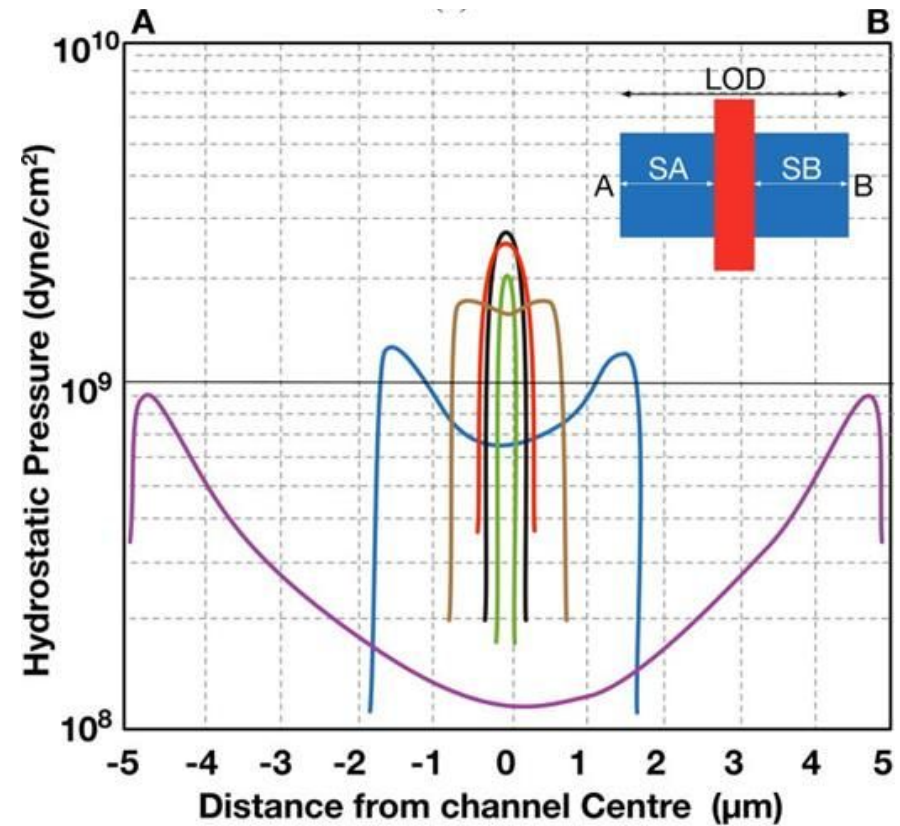


*SA/SB: Distance from active PC edge to active area edge from Source or Drain Side

Prior experience with Stress



(a)

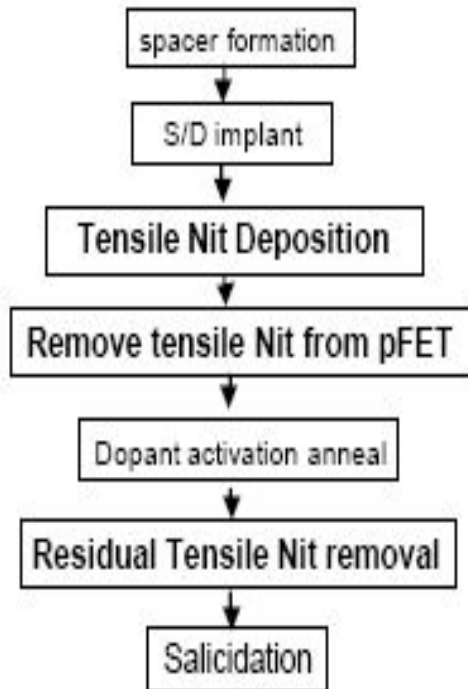


(b)

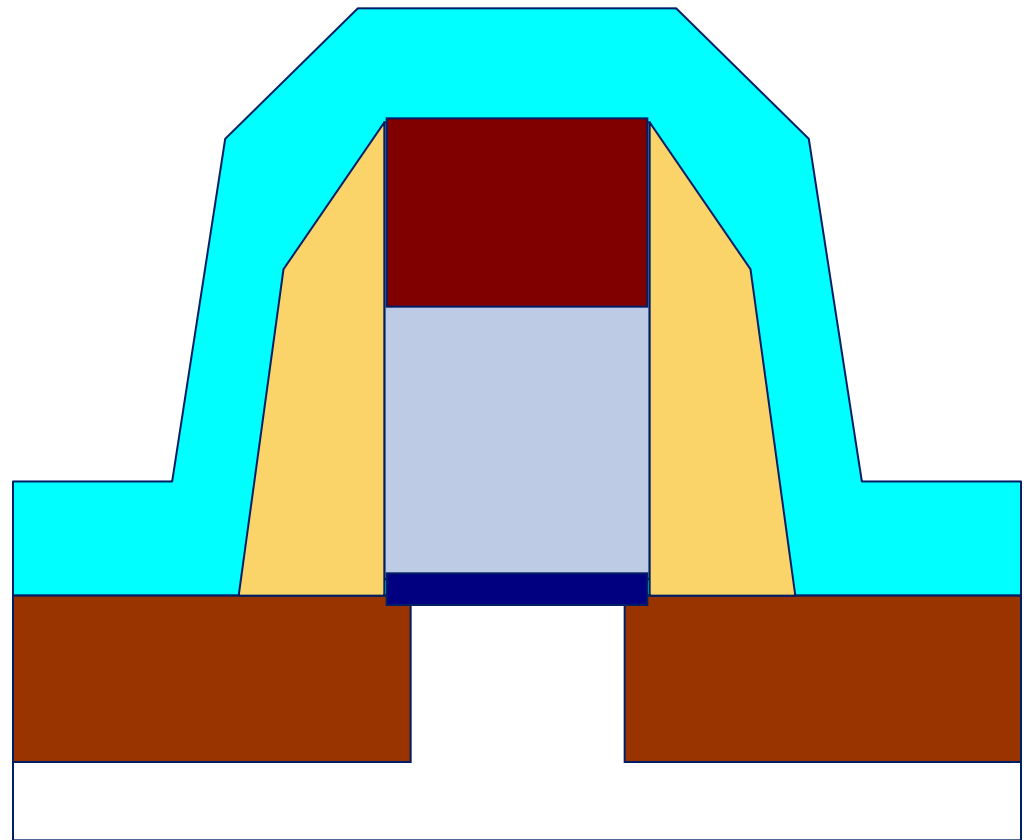
Stress in channel depends on STI fill (HARP or HDP), SA/SB and width of transistor

Rest of mobility enhancement techniques (SMT, dislocation and wafer/transport orientation)

SMT (Stress Memorization Technique)



When the SMT film is removed, stress from the film is still in the channel.



How does SMT work ?

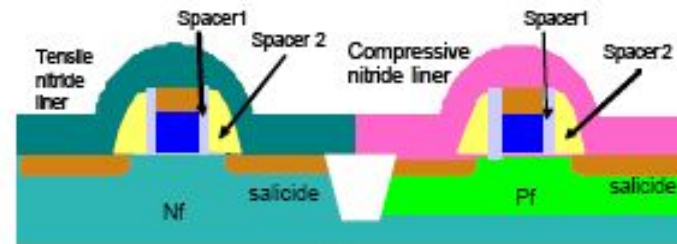
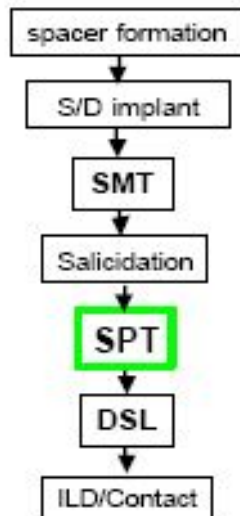
- Poly gate is amorphized with any additional amorphization implant or S/D implant
- With tensile capping layer, gate re-crystallizes during S/D RTA, resulting in compressive stress in poly gate due to difference in thermal expansion between Si and SiN. This compressive stress also induces tensile stress in channel
- This combination is beneficial for NFET while not so for PFET

110 direction on 001 surface

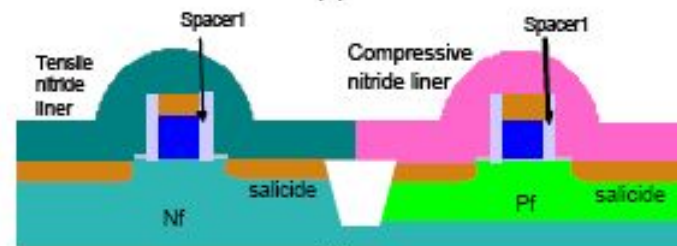
	nMOS	pMOS
Parallel stress	3.1%	-7.2%
Vertical stress	-5.3%	0.1%
Perpend. stress	1.7%	6.6%

Values are mobility change per 100MPa

SPT (Stress Proximity Technique)



(a)

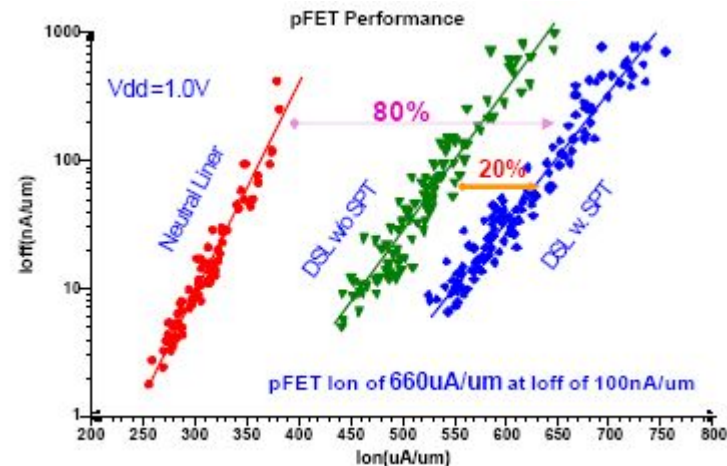


(b)

SPT is used to improve the PFET performance in nested poly pitch devices

Spacer2 is removed to bring the stress liner closer to the channel

Removal is done after the S/D step not to degrade the SCE



Edge dislocation stress

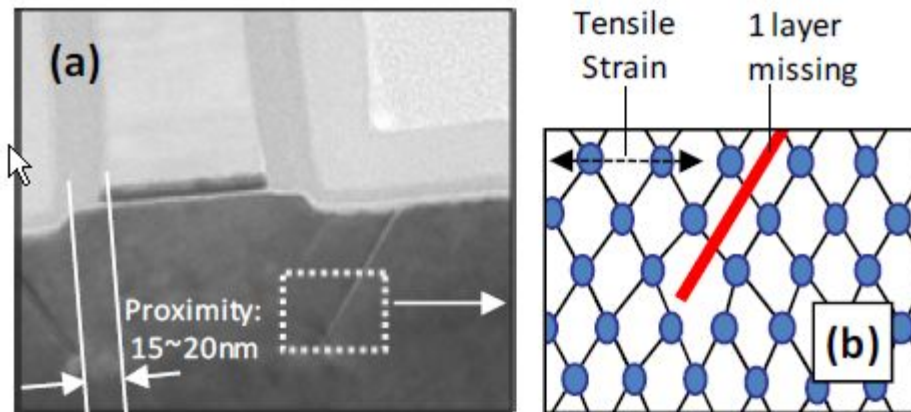
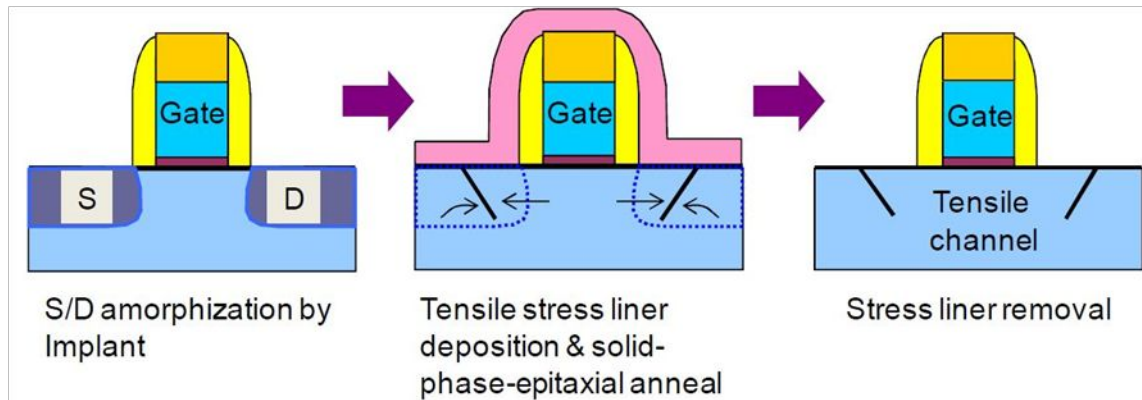
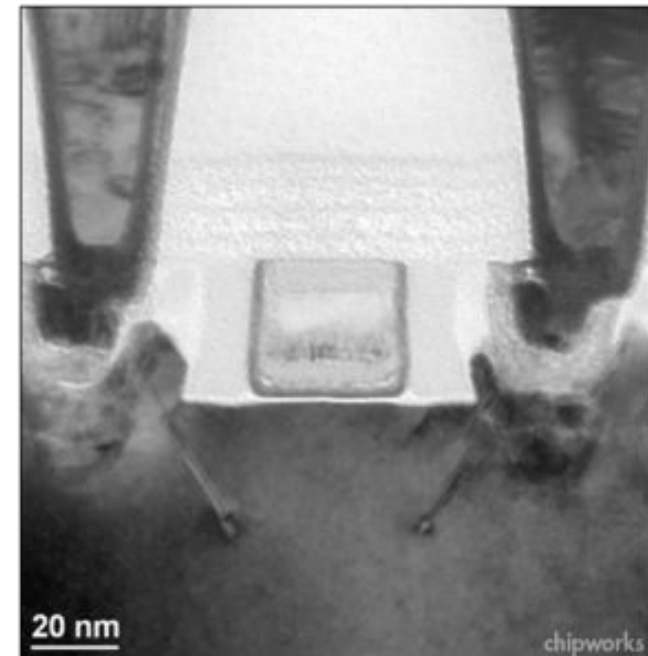
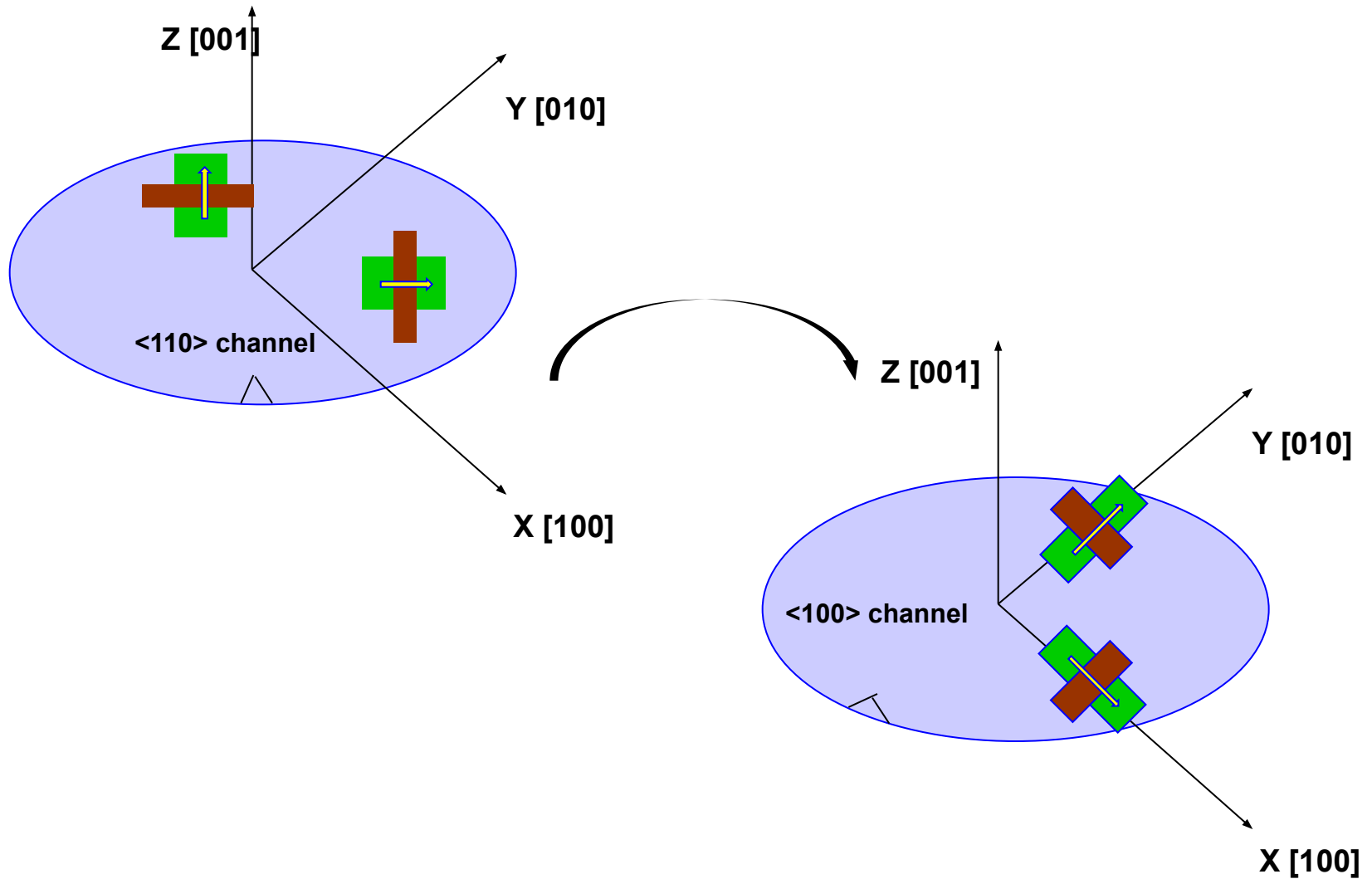


Fig. 2 S/D SMT induced mask-edge dislocation: (a) TEM image and (b) a schematic model showing stacking fault defect inside S/D region.



45 deg rotated wafer



Effect of rotation on mobility (<100> direction)

110 direction on 001 surface

	nMOS	pMOS
Parallel stress	3.1%	-7.2%
Vertical stress	-5.3%	0.1%
Perpend. stress	1.7%	6.6%

nMOS:

- effect of parallel and vertical stress same order

pMOS:

- effect of parallel stress dominant

100 direction on 001 surface

	nMOS	pMOS
Parallel stress	10.2%	-0.7%
Vertical stress	-5.3%	0.1%
Perpend. stress	-5.3 %	0.1%

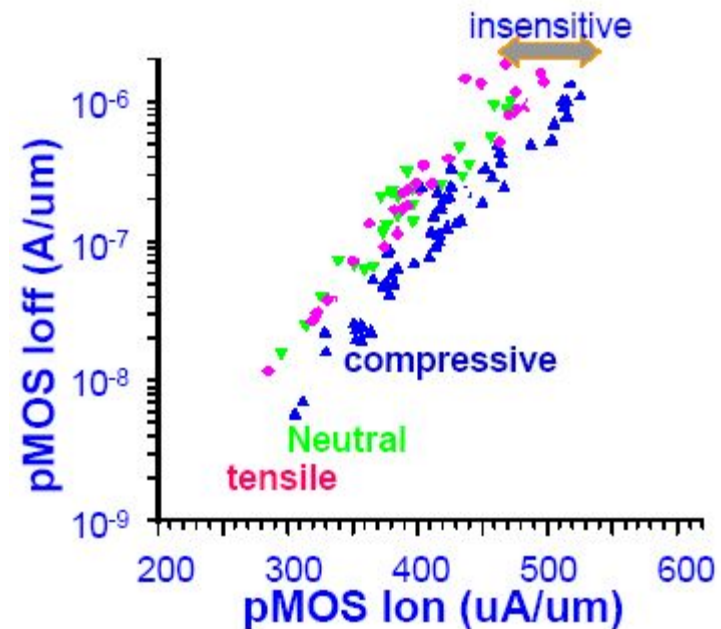
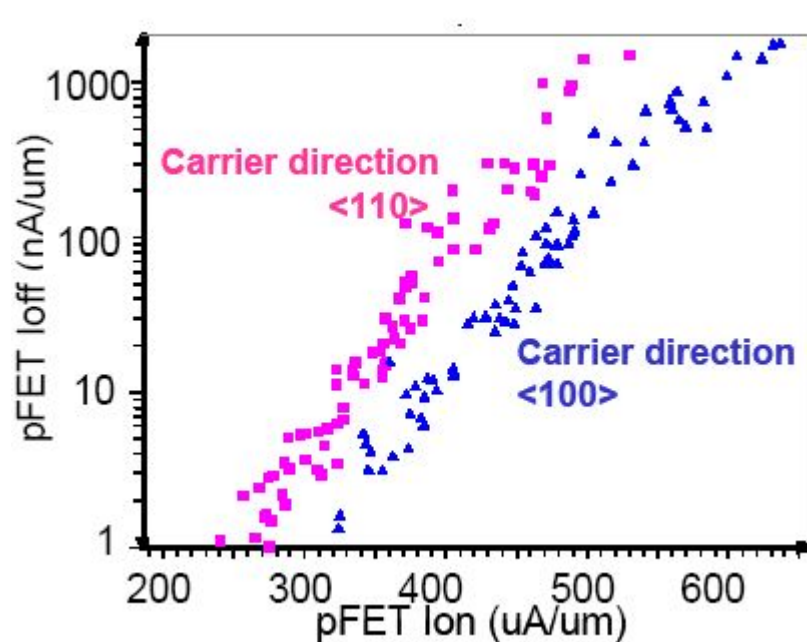
nMOS:

- large effect of parallel stress
- Perp. and vert. effect are the same

pMOS:

- Insensitive to every stress

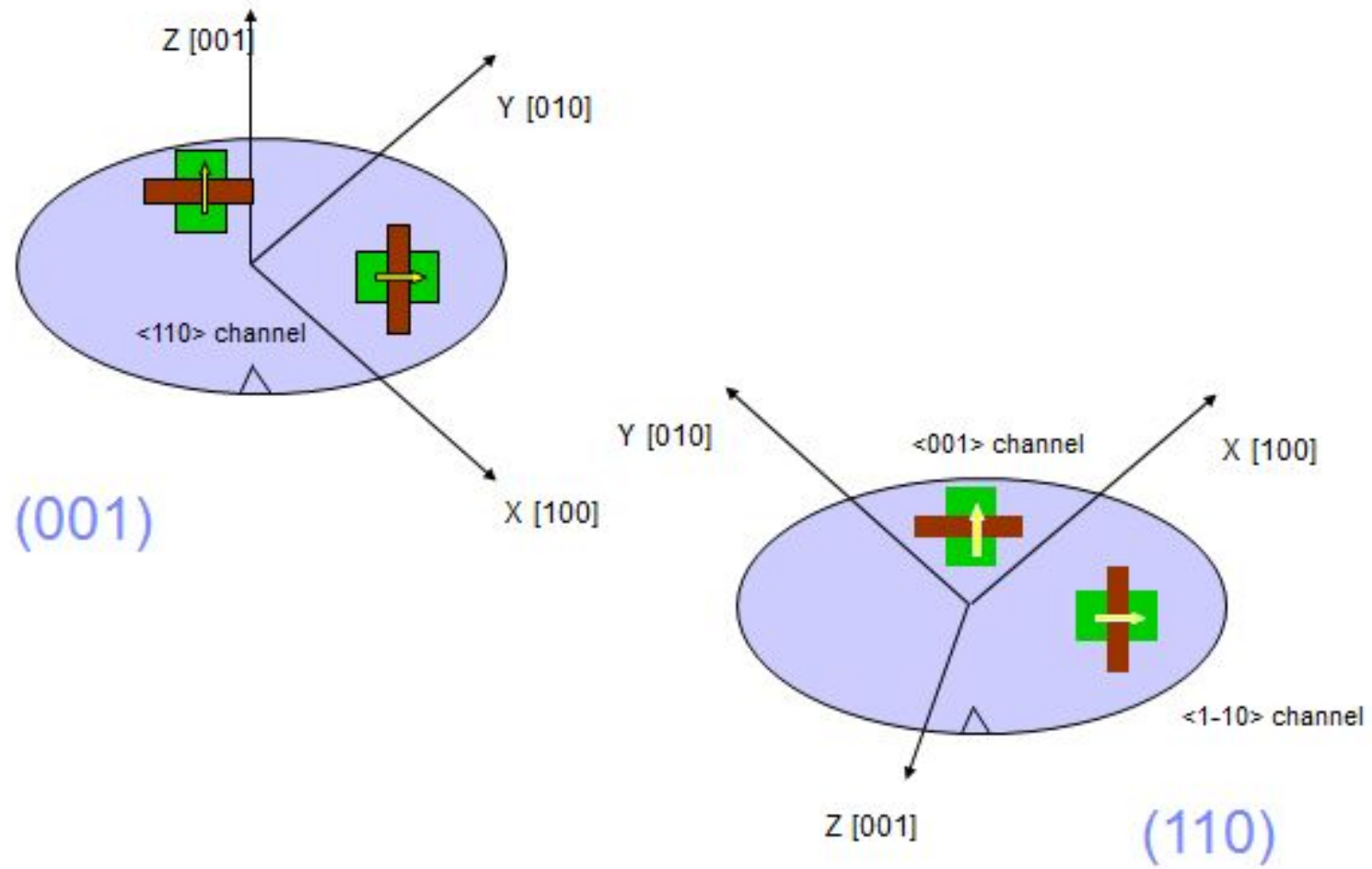
45 deg rotated wafer ($\langle 100 \rangle$ PFET)



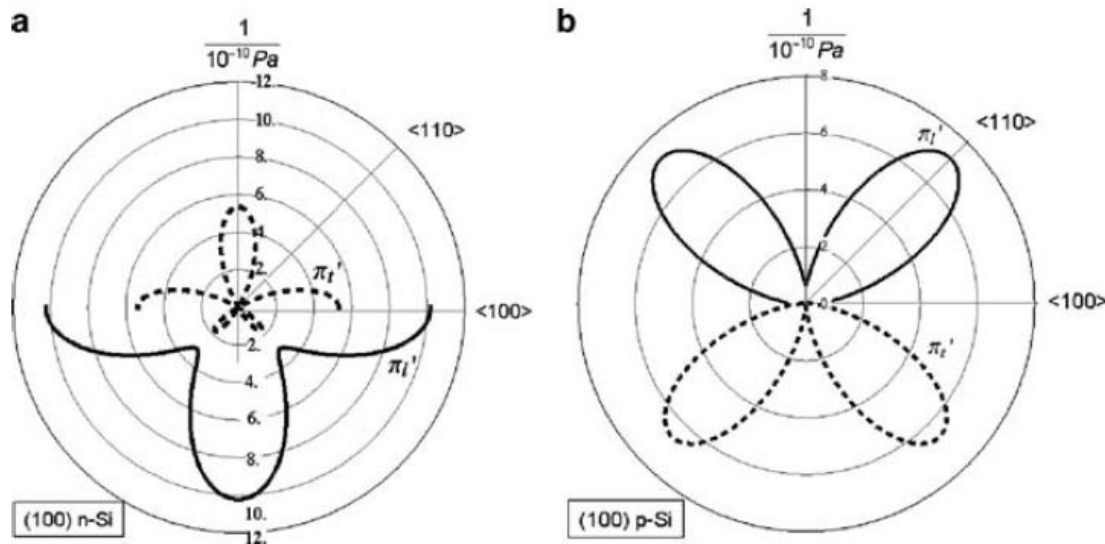
V. Chan, CICC 2006

- Improvement seen in short channel or narrow width devices
- Significant boost in PFET I_{on} without any added process steps
- PFET becomes insensitive to stress liner. Can save one mask by using one blanket tensile liner
- Used in 65, 45 and 32nm Low Power nodes (5X GIDL reduction on NFET)

Changing substrate orientation from (100) to (110)

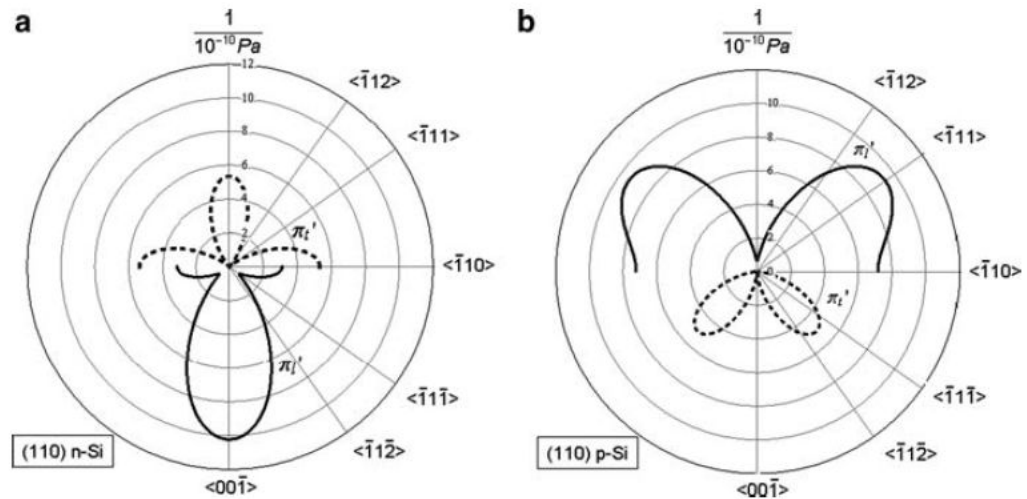


Piezoresistive coeffs for other orientation/direction



(001) Wafer

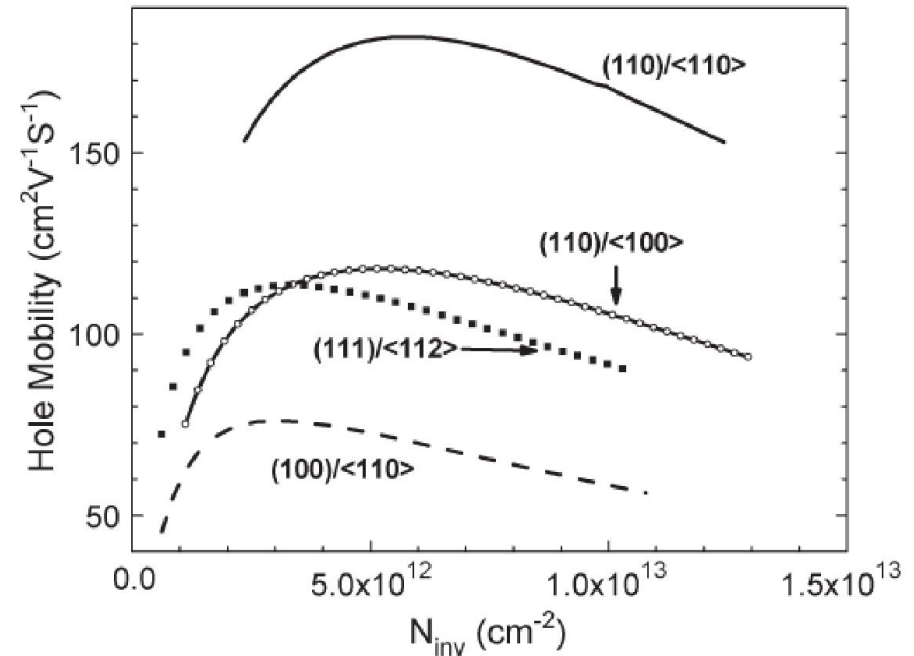
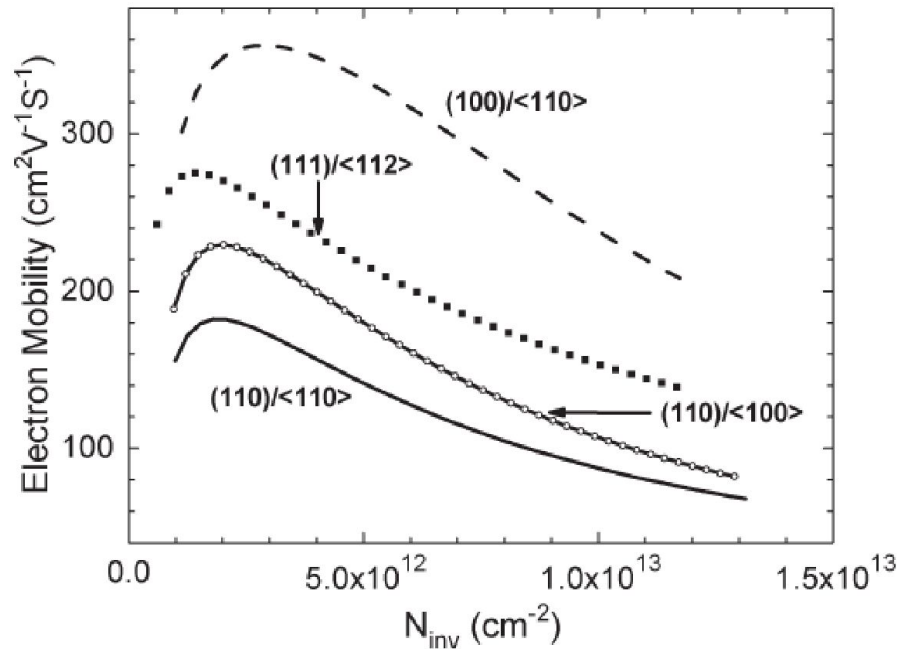
Fig. 8.5. Longitudinal and transverse π -coefficients for (a) n-type and (b) p-type Si in the (100) plane, i.e., $\theta = \angle 33' = 0$, and $\phi = \angle 11'$ from 0 to 180°



(110) Wafer

Fig. 8.8. Longitudinal and transverse π -coefficients for (a) n-type and (b) p-type Si in the (110) plane for ψ rotation from 0 to 180°

A closer look at mobility dependence on surface orientation and carrier flow direction

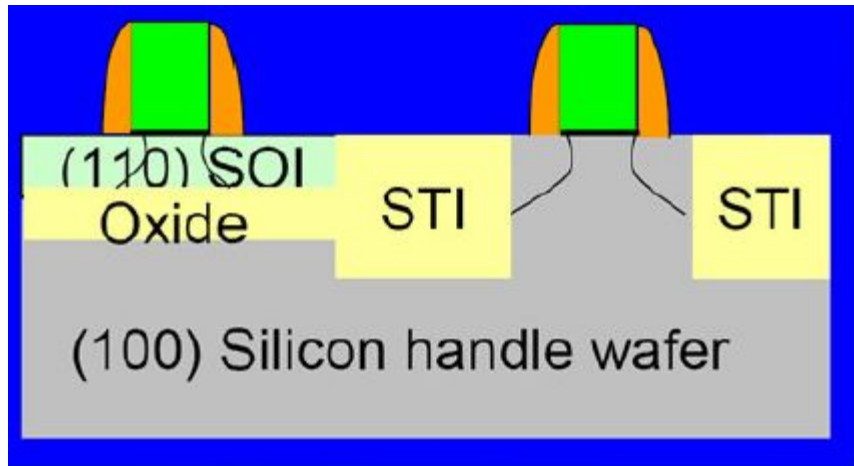


M. Yang, IEDM 2003

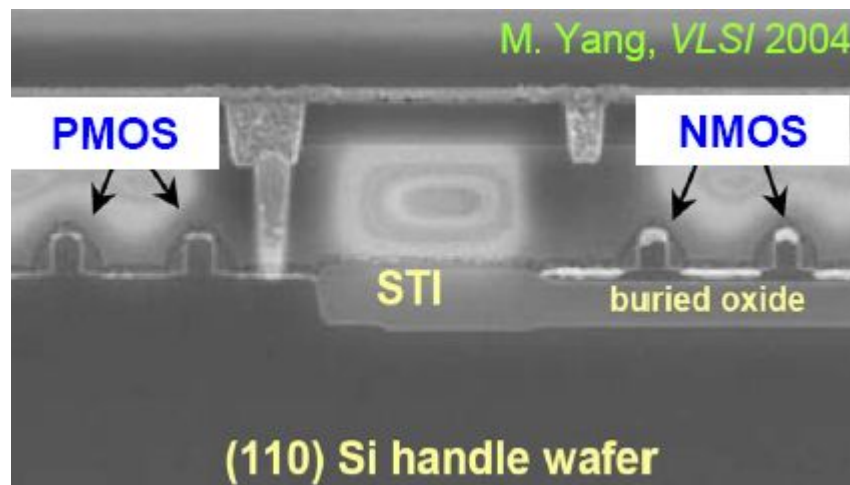
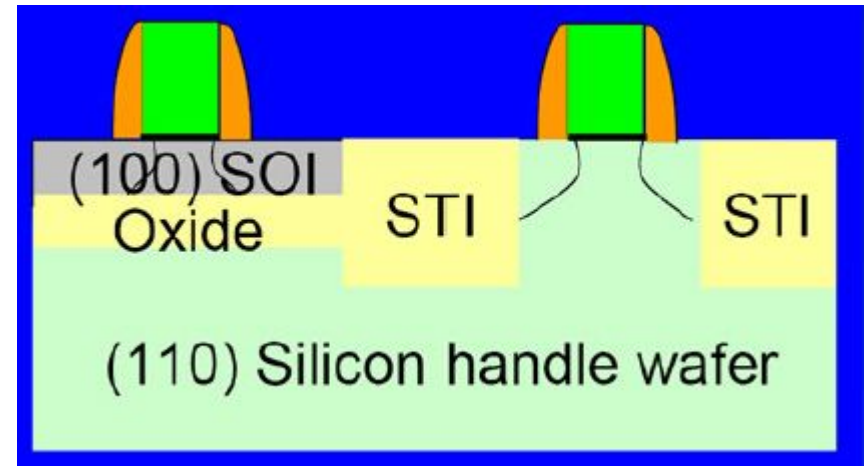
- Electron mobility is highest on (100) surface and hole mobility is highest on (110) surface
- Use hybrid orientation (100) for NFET and (110) for PFET
- On (110) substrate, mobility is dependent on the layout orientation

HOT (Hybrid Orientation technology)

PMOS on 110 NMOS on 100 epi



NMOS on 100 PMOS on 110



DSB (Direct Silicon Bonding)

